Supplementary Information

Sensitivity of Fuzzy-PSO Inertia Control to Linguistic Granularity and Rule-Base Structure

15 October 2026

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1 Fuzzy-controller configuration space

The main article shows representative panels. This section documents the full set of evaluated membership partitions, rule patterns, and induced control surfaces for all four granularities, $n \in \{3, 5, 7, 9\}$. The input and output partition families are I1 and O1 throughout.

1.1 Diversity input partition

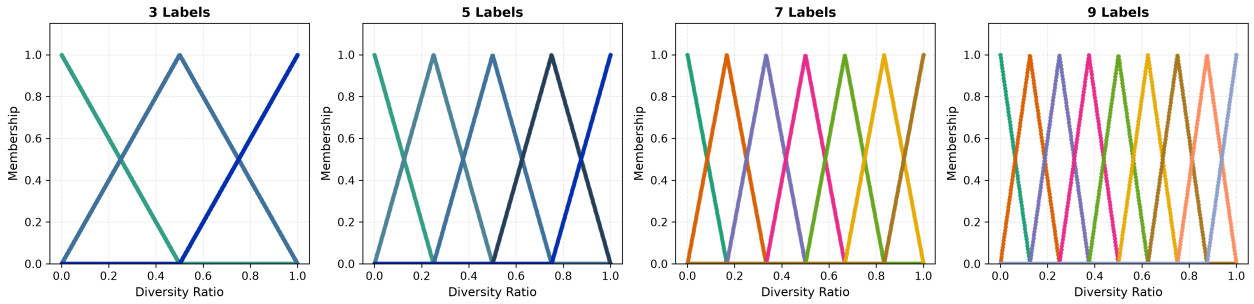


Figure 1: Diversity-input membership functions for all evaluated granularities.

1.2 Progress input partition

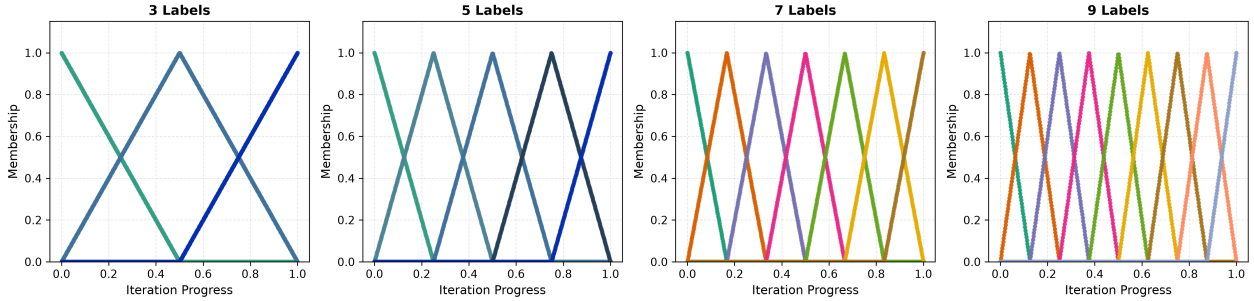


Figure 2: Progress-input membership functions for all evaluated granularities.

1.3 Inertia-weight output partition

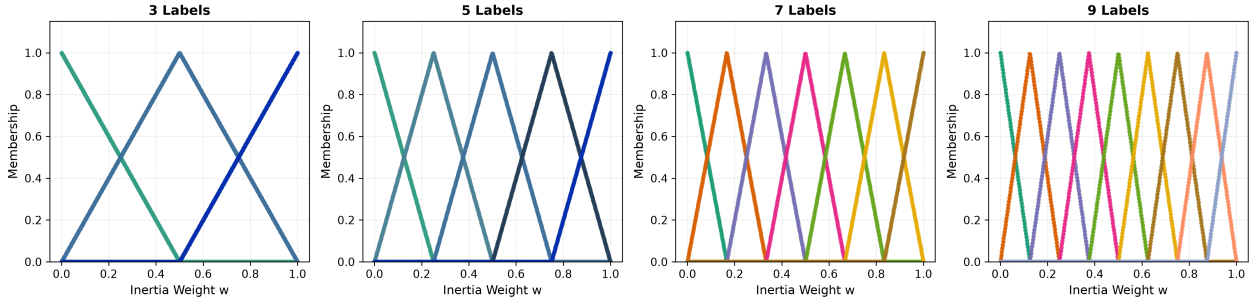


Figure 3: Output membership functions for the inertia weight, $w \in [0.1, 0.9]$, for all evaluated granularities.

1.4 Rule-base matrices

Each panel contains R1–R8 for the indicated granularity. Cell abbreviations and the shared colour scale are defined in the corresponding figure.

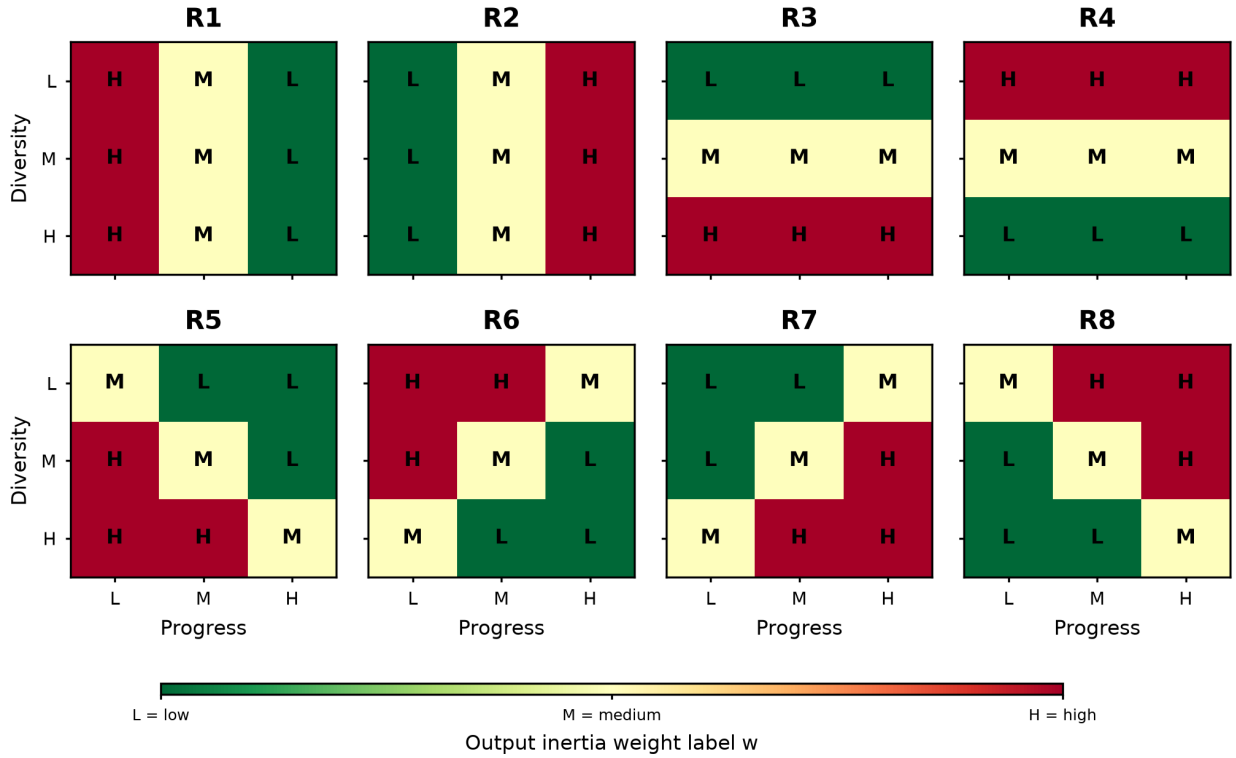


Figure 4: Rule-base matrix panel for $n = 3$.

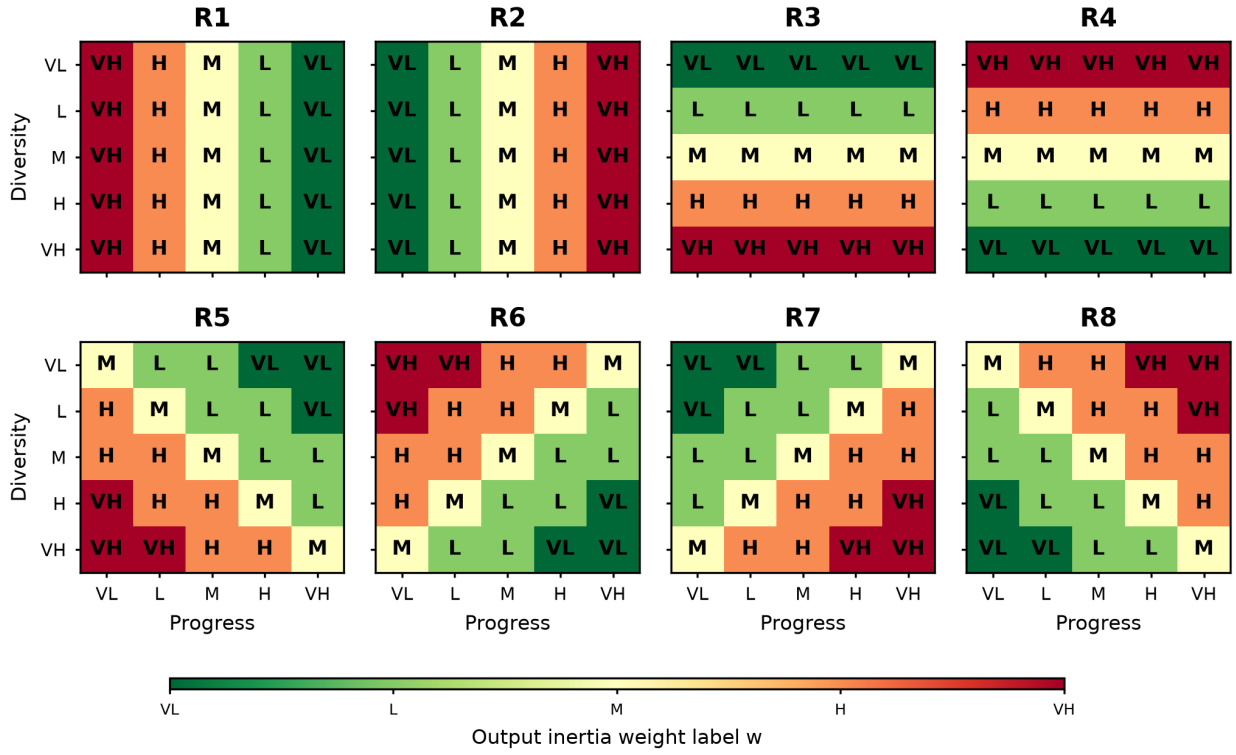


Figure 5: Rule-base matrix panel for $n = 5$.

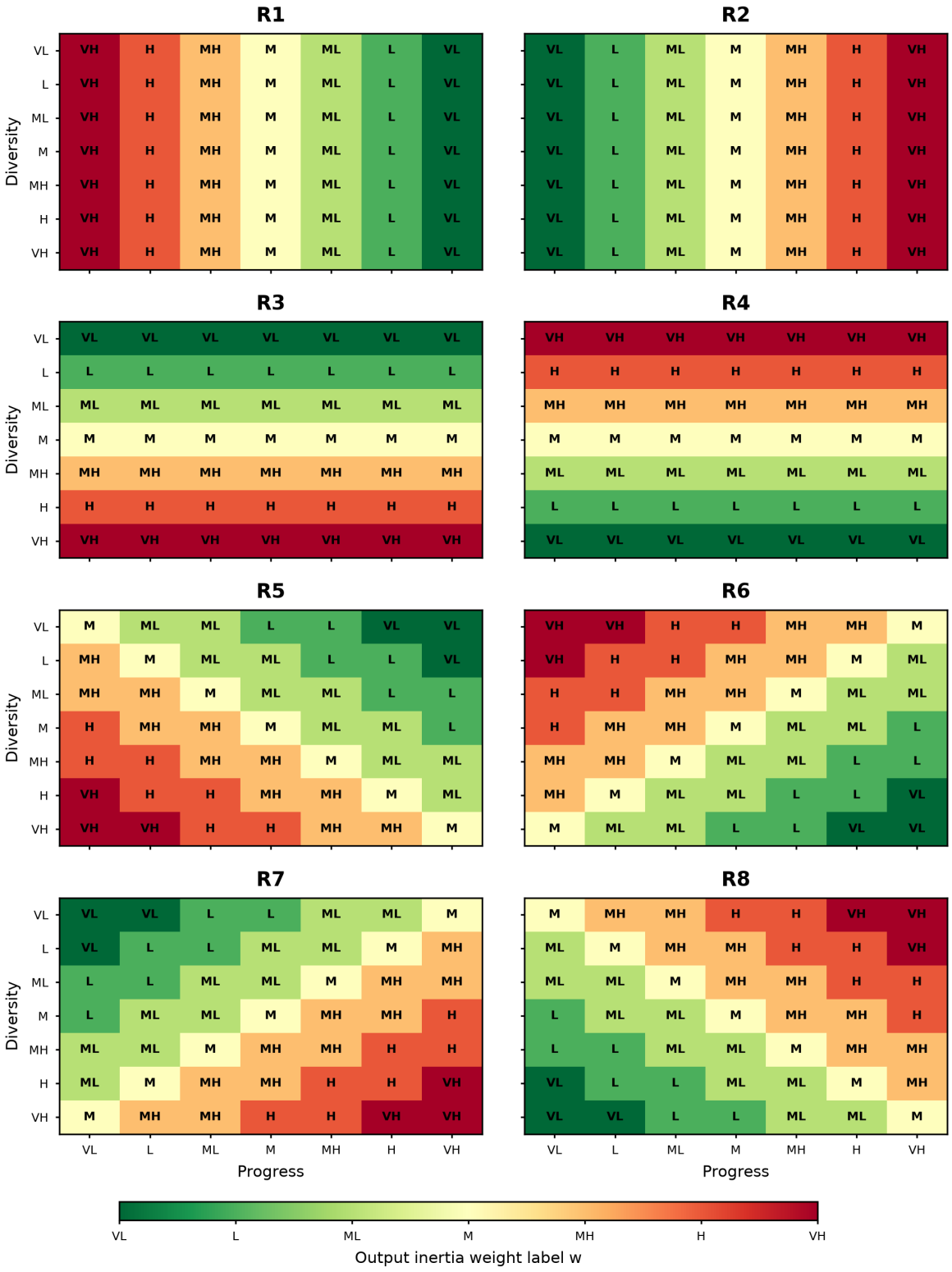


Figure 6: Rule-base matrix panel for $n = 7$.

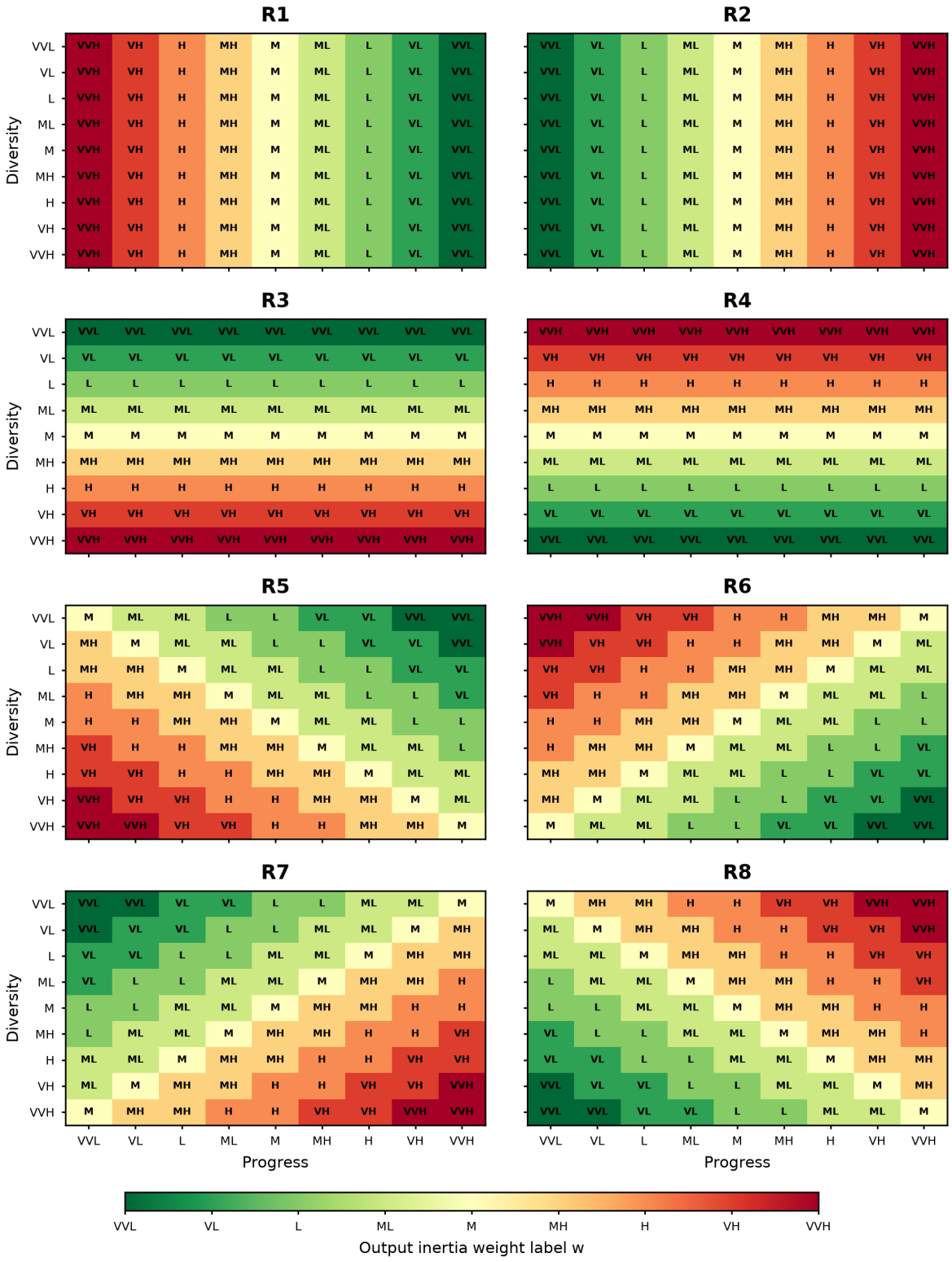


Figure 7: Rule-base matrix panel for $n = 9$.

1.5 Control surfaces

Each panel contains the eight rule-induced surfaces for the indicated granularity. Diversity and progress vary over $[0, 1]^2$; the output is the defuzzified inertia weight.

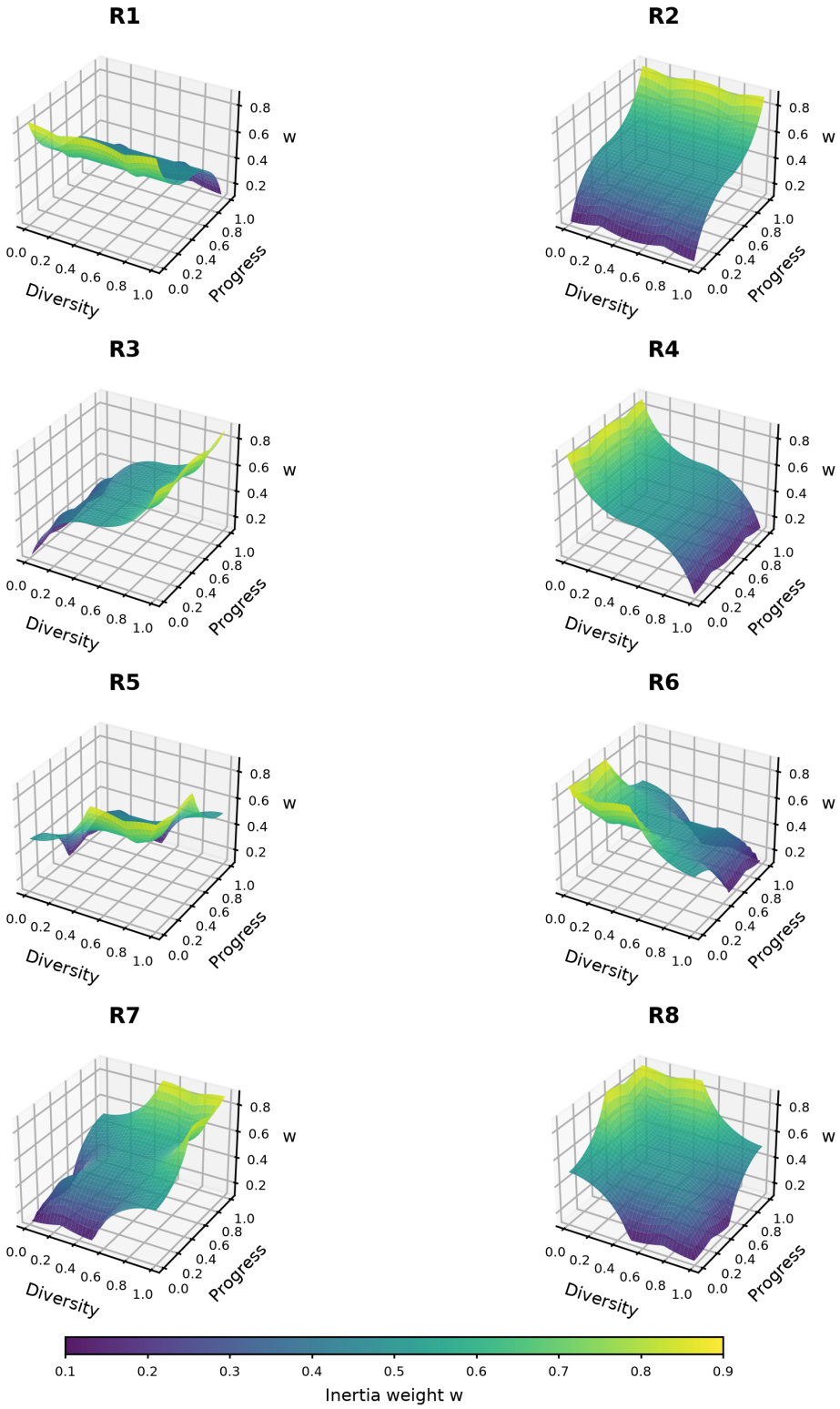


Figure 8: Control surfaces for R1–R8 at $n = 3$.

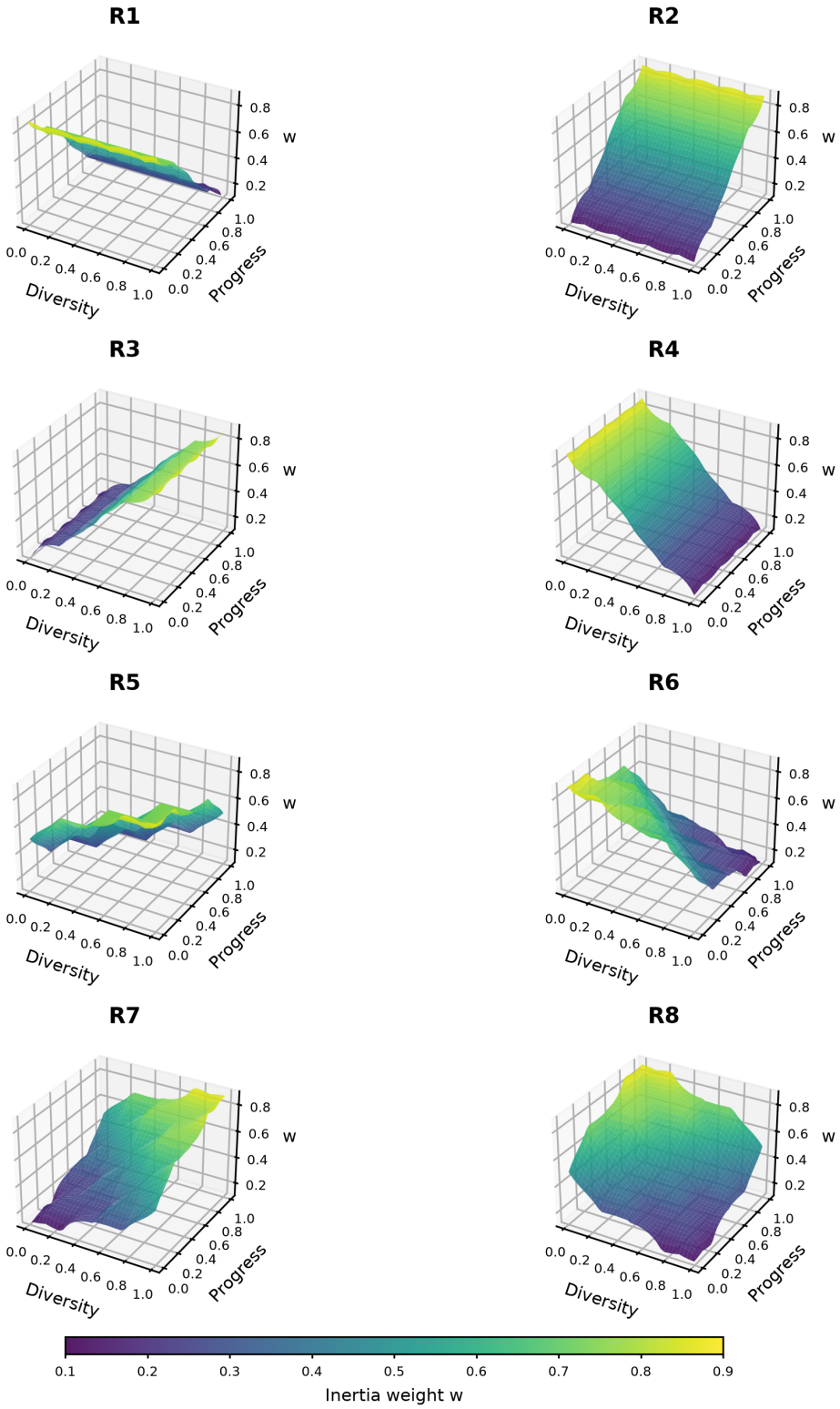


Figure 9: Control surfaces for R1–R8 at $n = 5$.

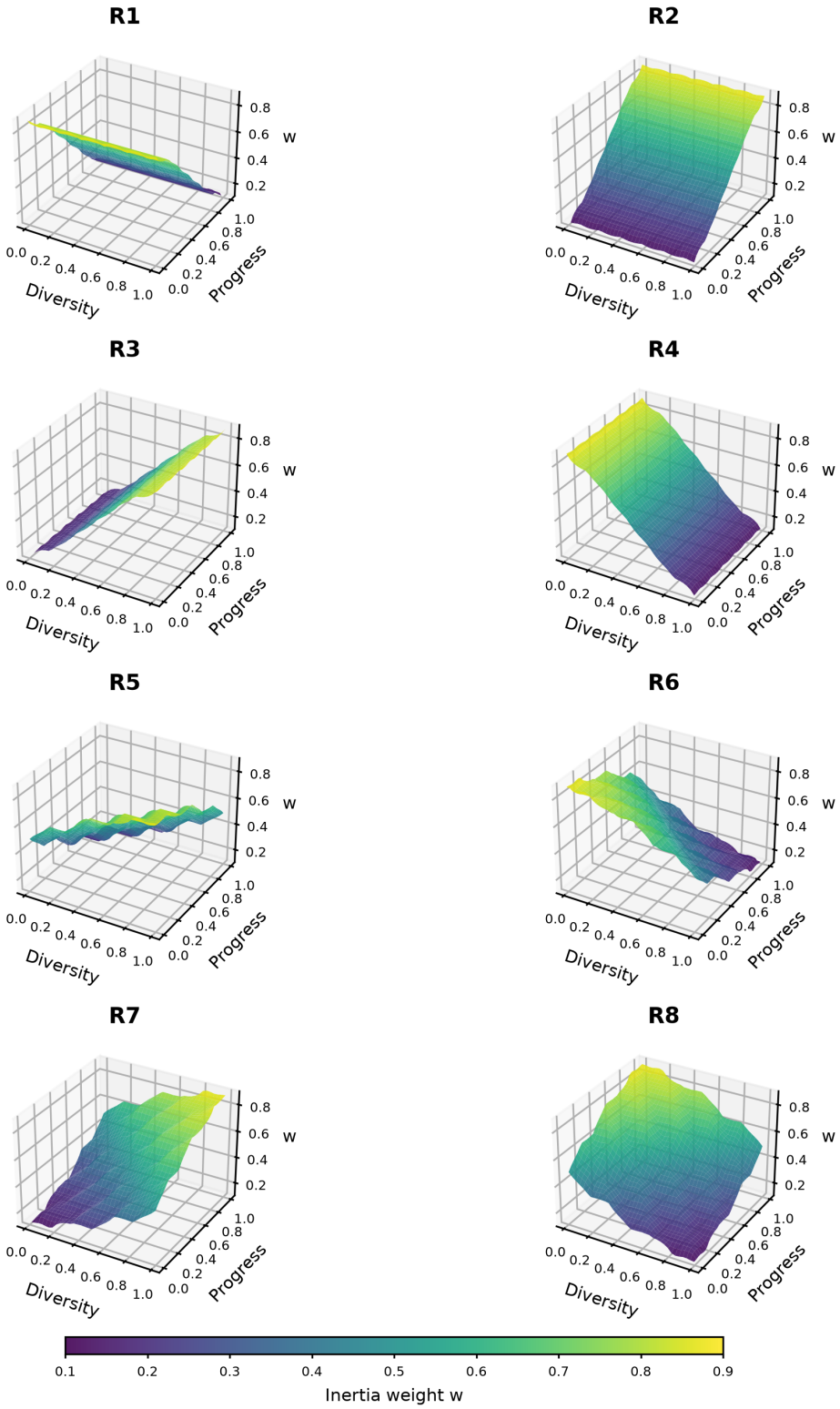


Figure 10: Control surfaces for R1–R8 at $n = 7$.

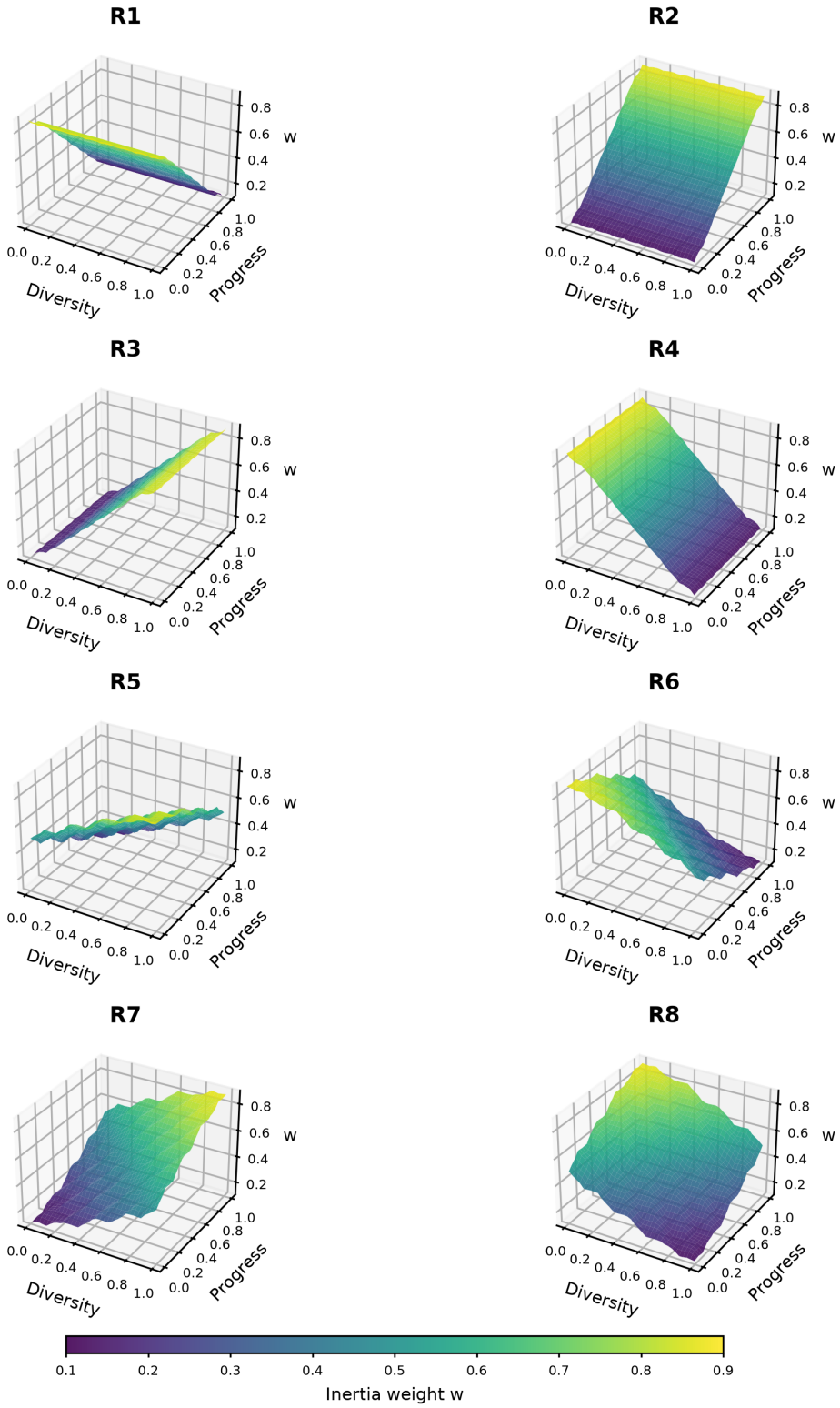


Figure 11: Control surfaces for R1–R8 at $n = 9$.

2 Complete function-level results

Table rows correspond to 62 function–dimension instances. Ranks use median normalized gap over all 33 configurations. Each p-value is from a two-sided Mann–Whitney U test on independent samples and is Bonferroni-adjusted across the 1,984 RQ4 comparisons. Since fitness is minimized, $A_{12} > 0.5$ favours the listed FCS configuration.

Table 1: Per-instance FCS–PSO results (part 1 of 3). Ranks use median normalized gap across 33 configurations.

Instance	Best FCS	n	R	PSO rk.	FCS rk.	PSO gap	FCS gap	p_{Bonf}	A_{12}
F1_D10	FCS-3-R1	3	R1	5	1	6.3×10^{-37}	1.4×10^{-44}	$<10^{-3}$	0.96
F1_D30	FCS-3-R1	3	R1	3	1	7.7×10^{-6}	1.7×10^{-9}	$<10^{-3}$	0.97
F1_D50	FCS-3-R1	3	R1	3	1	0.059	1.1×10^{-3}	$<10^{-3}$	0.99
F1_D100	FCS-3-R1	3	R1	2	1	16.744	10.263	1.000	0.72
F2_D10	FCS-3-R1	3	R1	6	1	3.5×10^{-18}	7.7×10^{-25}	$<10^{-3}$	0.98
F2_D30	FCS-3-R1	3	R1	2	1	4.5×10^{-4}	5.2×10^{-6}	$<10^{-3}$	0.98
F2_D50	FCS-3-R1	3	R1	2	1	0.099	8×10^{-3}	$<10^{-3}$	0.98
F2_D100	FCS-3-R1	3	R1	2	1	2.357	1.375	$<10^{-3}$	0.94
F3_D10	FCS-5-R5	5	R5	9	1	8.3×10^{-11}	2.8×10^{-15}	$<10^{-3}$	0.93
F3_D30	FCS-7-R5	7	R5	3	1	41.57	37.246	1.000	0.58
F3_D50	FCS-5-R7	5	R7	13	1	1692.014	1247.305	1.000	0.65
F3_D100	FCS-9-R7	9	R7	18	1	2×10^{4}	1.4×10^{4}	0.027	0.82
F4_D10	FCS-9-R5	9	R5	5	1	1.4×10^{-8}	3.2×10^{-11}	$<10^{-3}$	0.90
F4_D30	FCS-9-R1	9	R1	6	1	1.379	1.281	1.000	0.56
F4_D50	FCS-5-R1	5	R1	3	1	4.347	4.234	1.000	0.56
F4_D100	FCS-5-R5	5	R5	1	2	14.314	14.356	1.000	0.50
F5_D10	FCS-3-R1	3	R1	5	1	4.768	4.197	0.215	0.79
F5_D30	FCS-3-R1	3	R1	4	1	29.198	27.947	1.000	0.50
F5_D50	FCS-3-R1	3	R1	2	1	150.43	116.51	1.000	0.61
F5_D100	FCS-3-R1	3	R1	2	1	955.011	814.72	1.000	0.67
F6_D10	FCS-3-R1	3	R1	4	1	2.8×10^{-32}	0	0.006	0.82

p_{Bonf} is the Bonferroni-adjusted two-sided independent-sample Mann–Whitney U p-value across 1,984 RQ4 tests. $A_{12} > 0.5$ favours the listed FCS because lower fitness is better. Parentheses report configurations sharing the best median-gap rank.

Table 2: Per-instance FCS–PSO results (part 2 of 3). Ranks use median normalized gap across 33 configurations.

Instance	Best FCS	n	R	PSO rk.	FCS rk.	PSO gap	FCS gap	p_{Bonf}	A_{12}
F6_D30	FCS-3-R1	3	R1	2	1	2.4×10^{-6}	1.9×10^{-9}	$<10^{-3}$	0.99
F6_D50	FCS-3-R1	3	R1	2	1	0.046	1.1×10^{-3}	$<10^{-3}$	0.97
F6_D100	FCS-3-R1	3	R1	2	1	22.053	7.284	$<10^{-3}$	0.93
F7_D10	FCS-7-R6	7	R6	8	1	1.3×10^{-3}	1×10^{-3}	1.000	0.56
F7_D30	FCS-5-R1	5	R1	1	2	9×10^{-3}	9.1×10^{-3}	1.000	0.52
F7_D50	FCS-3-R1	3	R1	2	1	0.028	0.027	1.000	0.53
F7_D100	FCS-5-R1	5	R1	4	1	0.142	0.132	1.000	0.55
F8_D10	FCS-9-R8	9	R8	6	1	0.415	0.396	1.000	0.56
F8_D30	FCS-7-R3	7	R3	12	1	0.495	0.465	1.000	0.55
F8_D50	FCS-3-R1	3	R1	1	2	0.483	0.491	1.000	0.43
F8_D100	FCS-5-R6	5	R6	1	2	0.502	0.503	1.000	0.54
F9_D10	FCS-5-R2	5	R2	11	1	10.945	8.955	1.000	0.59
F9_D30	FCS-5-R5	5	R5	7	1	35.818	31.853	1.000	0.56
F9_D50	FCS-3-R5	3	R5	14	1	67.66	55.771	1.000	0.68
F9_D100	FCS-7-R5	7	R5	4	1	132.303	122.623	1.000	0.61
F10_D10	FCS-3-R1 (6 tied)	3	R1	4	4	4×10^{-15}	4×10^{-15}	1.000	0.45
F10_D30	FCS-3-R1	3	R1	2	1	6.6×10^{-4}	1.5×10^{-5}	$<10^{-3}$	0.93
F10_D50	FCS-3-R1	3	R1	4	1	0.522	0.019	0.094	0.80
F10_D100	FCS-3-R1	3	R1	3	1	2.192	1.966	1.000	0.74
F11_D10	FCS-9-R1	9	R1	10	1	0.091	0.069	1.000	0.64
F11_D30	FCS-7-R1	7	R1	5	1	7.5×10^{-3}	1.4×10^{-3}	1.000	0.54

p_{Bonf} is the Bonferroni-adjusted two-sided independent-sample Mann–Whitney U p-value across 1,984 RQ4 tests. $A_{12} > 0.5$ favours the listed FCS because lower fitness is better. Parentheses report configurations sharing the best median-gap rank.

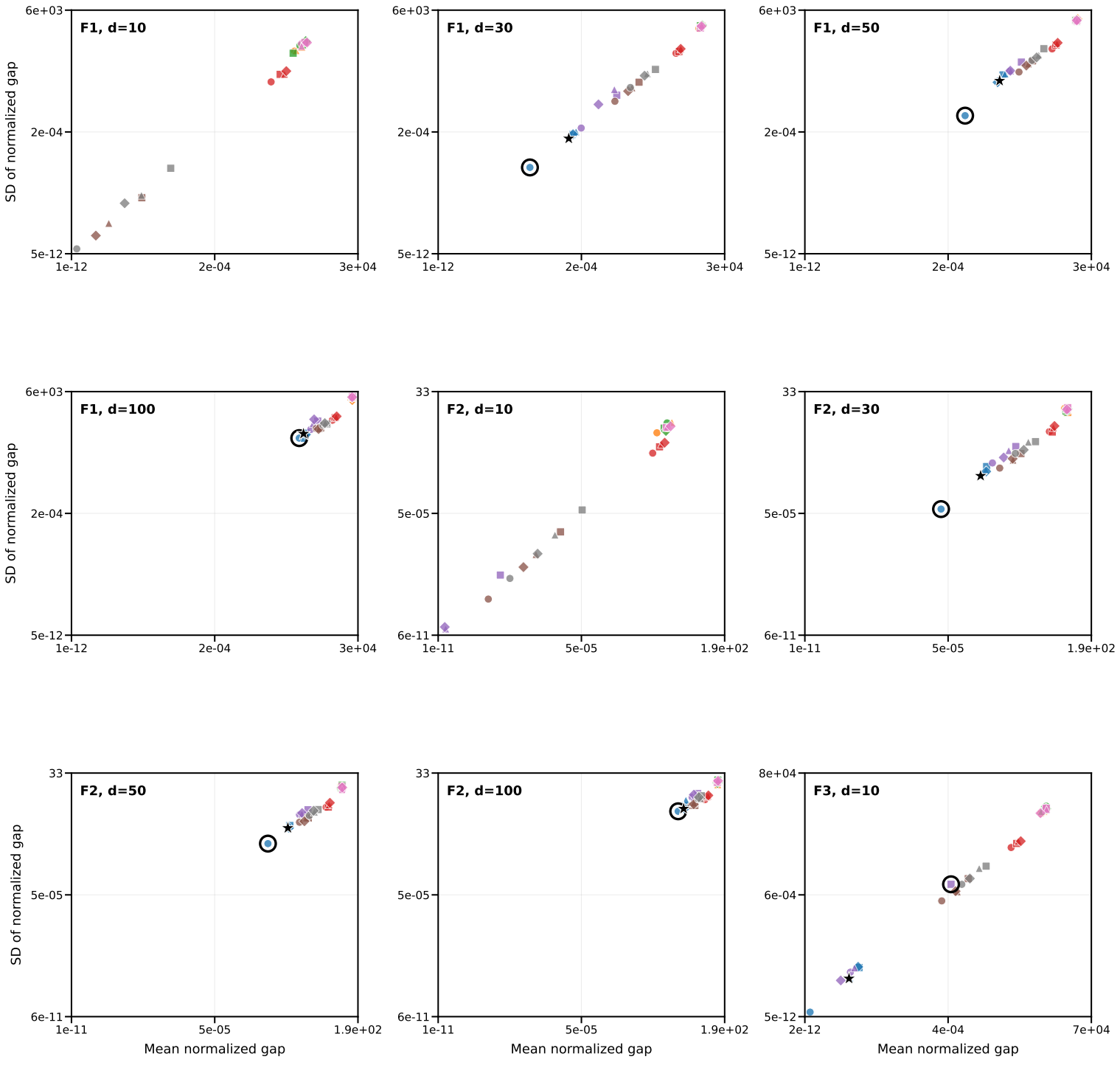
Table 3: Per-instance FCS–PSO results (part 3 of 3). Ranks use median normalized gap across 33 configurations.

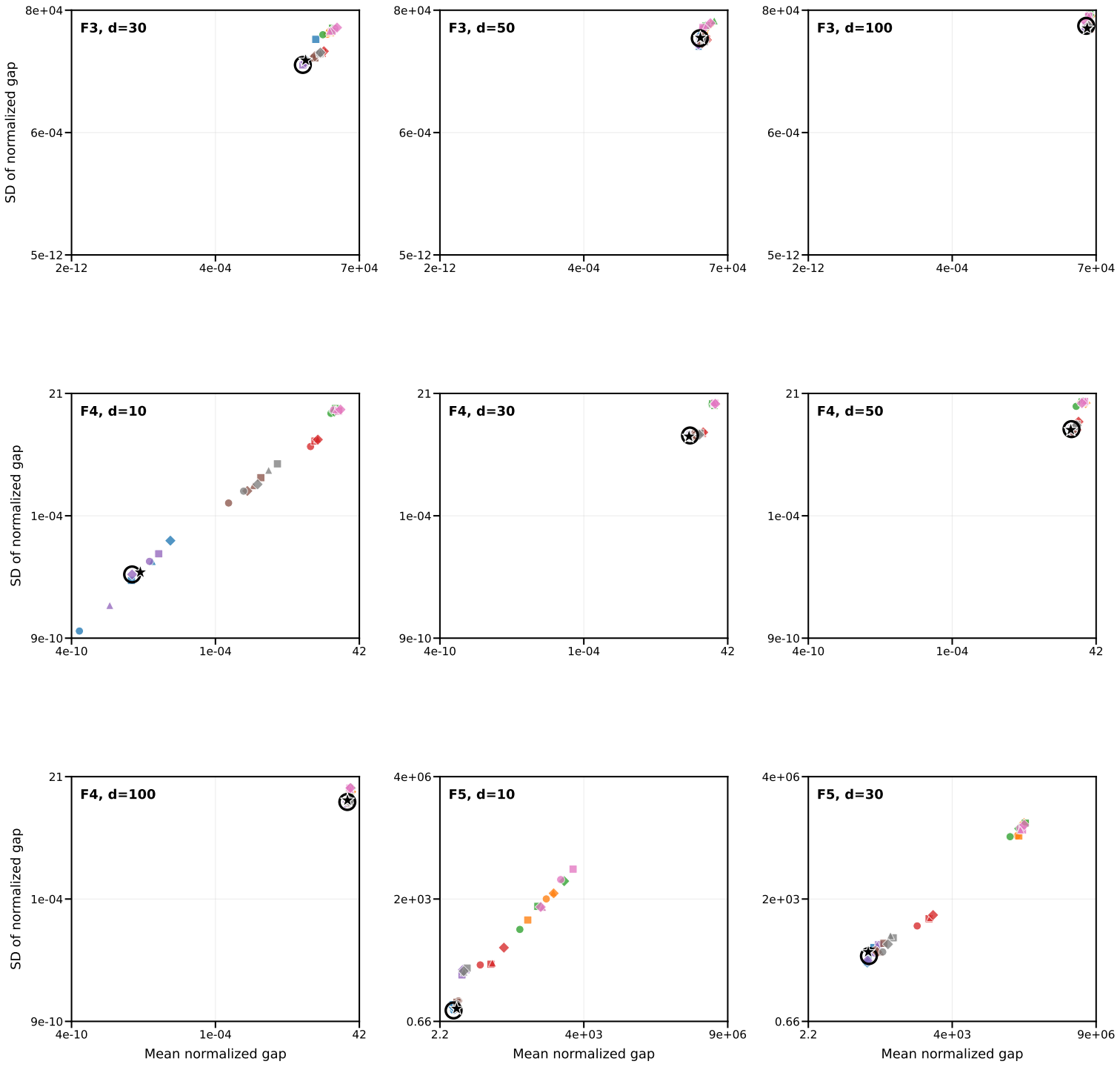
Instance	Best FCS	n	R	PSO rk.	FCS rk.	PSO gap	FCS gap	p_{Bonf}	A_{12}
F11_D50	FCS-3-R1	3	R1	3	1	0.064	0.012	$< 10^{-3}$	0.91
F11_D100	FCS-3-R1	3	R1	2	1	1.168	1.042	0.004	0.85
F12_D10	FCS-3-R1	3	R1	2	1	4.8×10^{-32}	4.7×10^{-32}	0.284	0.77
F12_D30	FCS-3-R1	3	R1	2	1	1.2×10^{-8}	2.1×10^{-11}	0.065	0.81
F12_D50	FCS-7-R1	7	R1	1	2	4.1×10^{-4}	4.9×10^{-4}	1.000	0.49
F12_D100	FCS-5-R1	5	R1	3	1	2.769	2.313	1.000	0.67
F13_D10	FCS-3-R1	3	R1	3	1	1.8×10^{-32}	1.3×10^{-32}	0.126	0.76
F13_D30	FCS-3-R1	3	R1	2	1	2.4×10^{-6}	9.4×10^{-9}	1.000	0.65
F13_D50	FCS-3-R1	3	R1	3	1	0.025	8.9×10^{-3}	0.112	0.80
F13_D100	FCS-3-R1	3	R1	3	1	16.942	9.312	1.000	0.73
F14_D2	FCS-3-R1 (6 tied)	3	R1	27	3	2.968	0.992	1.000	0.58
F15_D4	FCS-3-R1	3	R1	11	1	7.5×10^{-6}	7.5×10^{-6}	1.000	0.75
F16_D2	FCS-3-R1 (32 tied)	3	R1	17	17	0	0	1.000	0.56
F17_D2	FCS-3-R1 (32 tied)	3	R1	17	17	0	0	1.000	0.50
F18_D2	FCS-3-R1 (28 tied)	3	R1	15	15	0	0	1.000	0.45
F19_D3	FCS-3-R1 (32 tied)	3	R1	17	17	0	0	1.000	0.56
F20_D6	FCS-3-R1 (27 tied)	3	R1	14	14	0	0	1.000	0.55
F21_D4	FCS-7-R1	7	R1	24	1	0.736	0	1.000	0.61
F22_D4	FCS-3-R6 (2 tied)	3	R6	20	1	0.642	0	1.000	0.57
F23_D4	FCS-5-R6	5	R6	1	2	0	1.7×10^{-16}	1.000	0.34

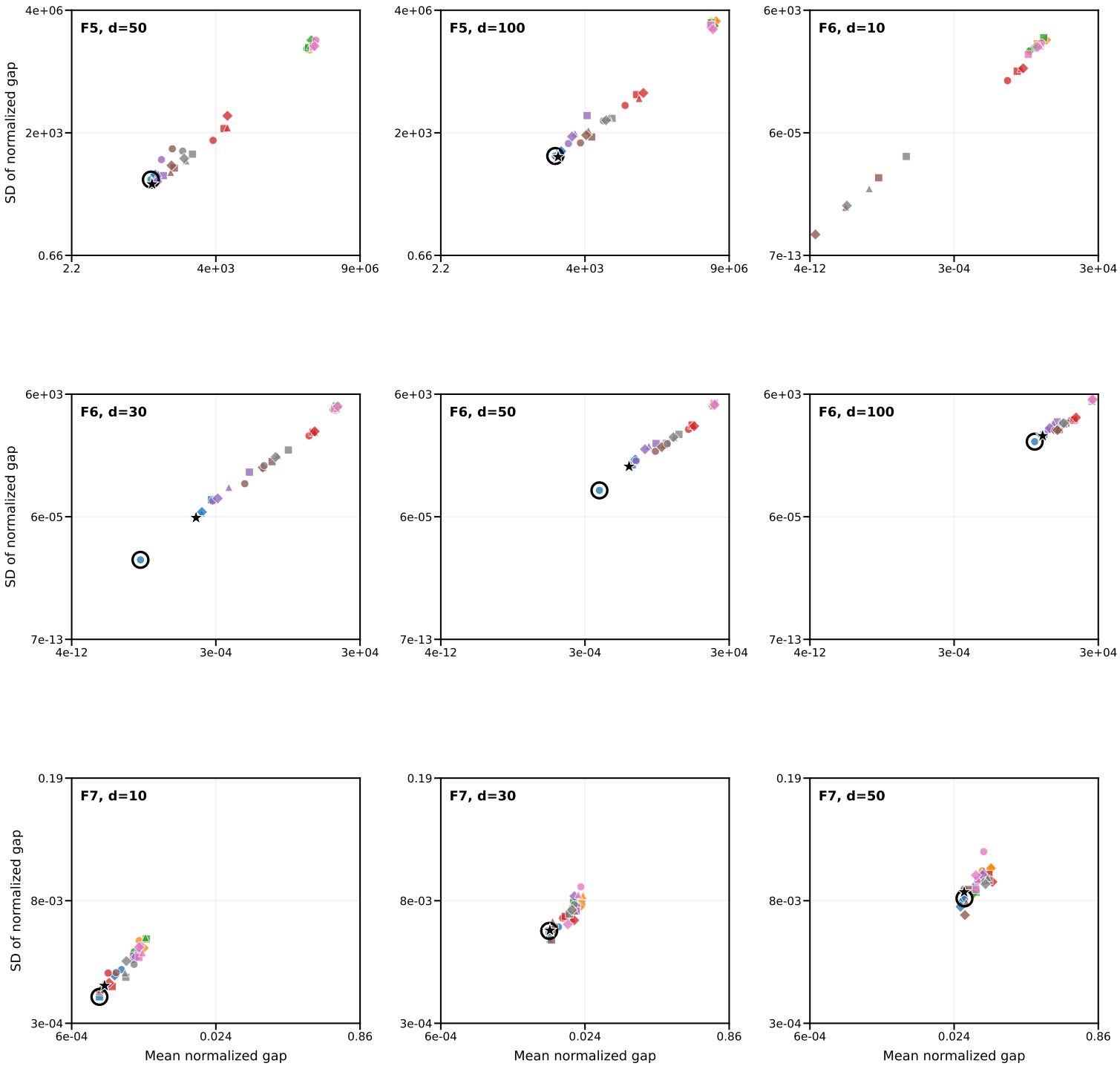
p_{Bonf} is the Bonferroni-adjusted two-sided independent-sample Mann–Whitney U p-value across 1,984 RQ4 tests. $A_{12} > 0.5$ favours the listed FCS because lower fitness is better. Parentheses report configurations sharing the best median-gap rank.

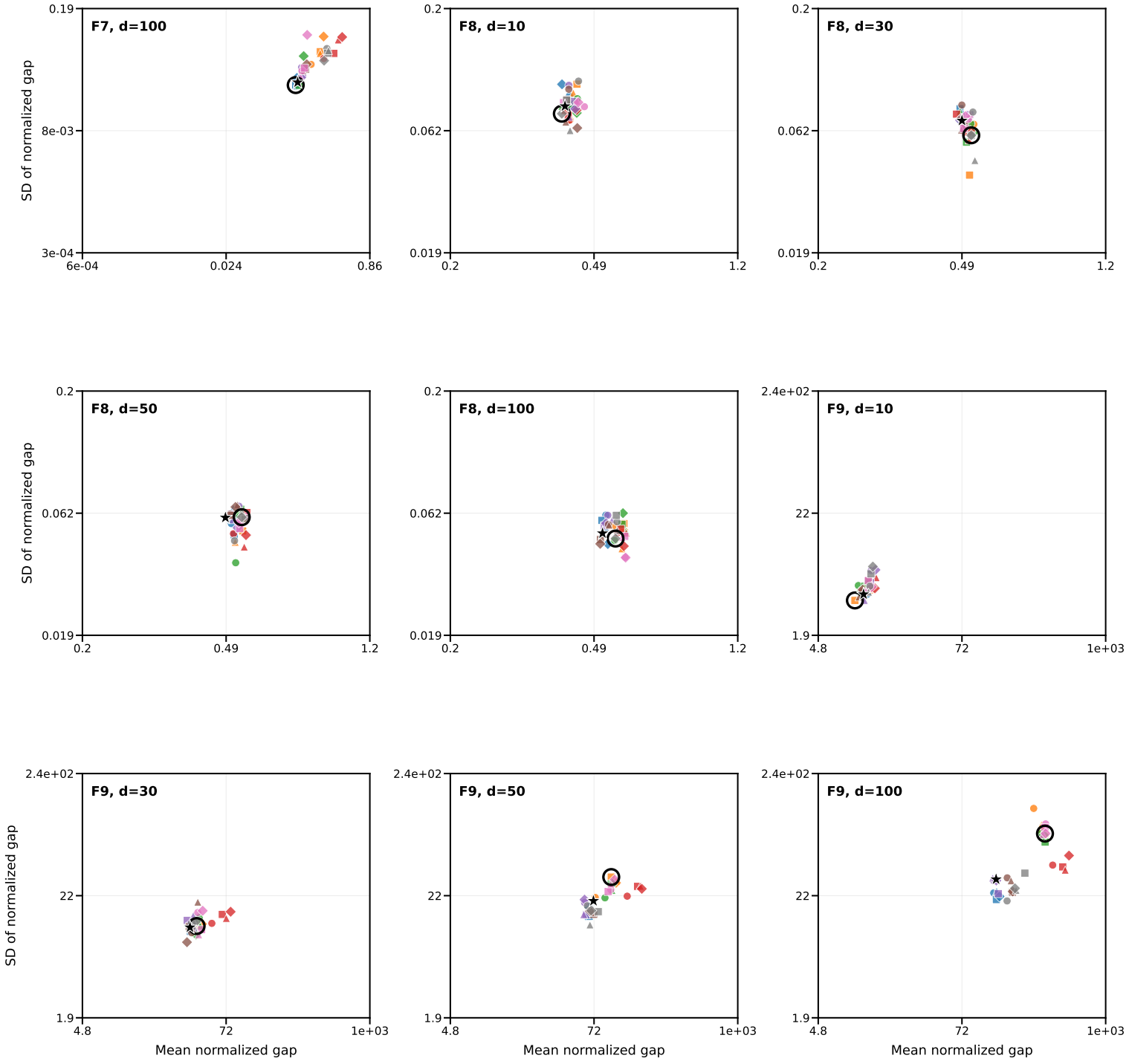
3 Performance versus run-to-run dispersion

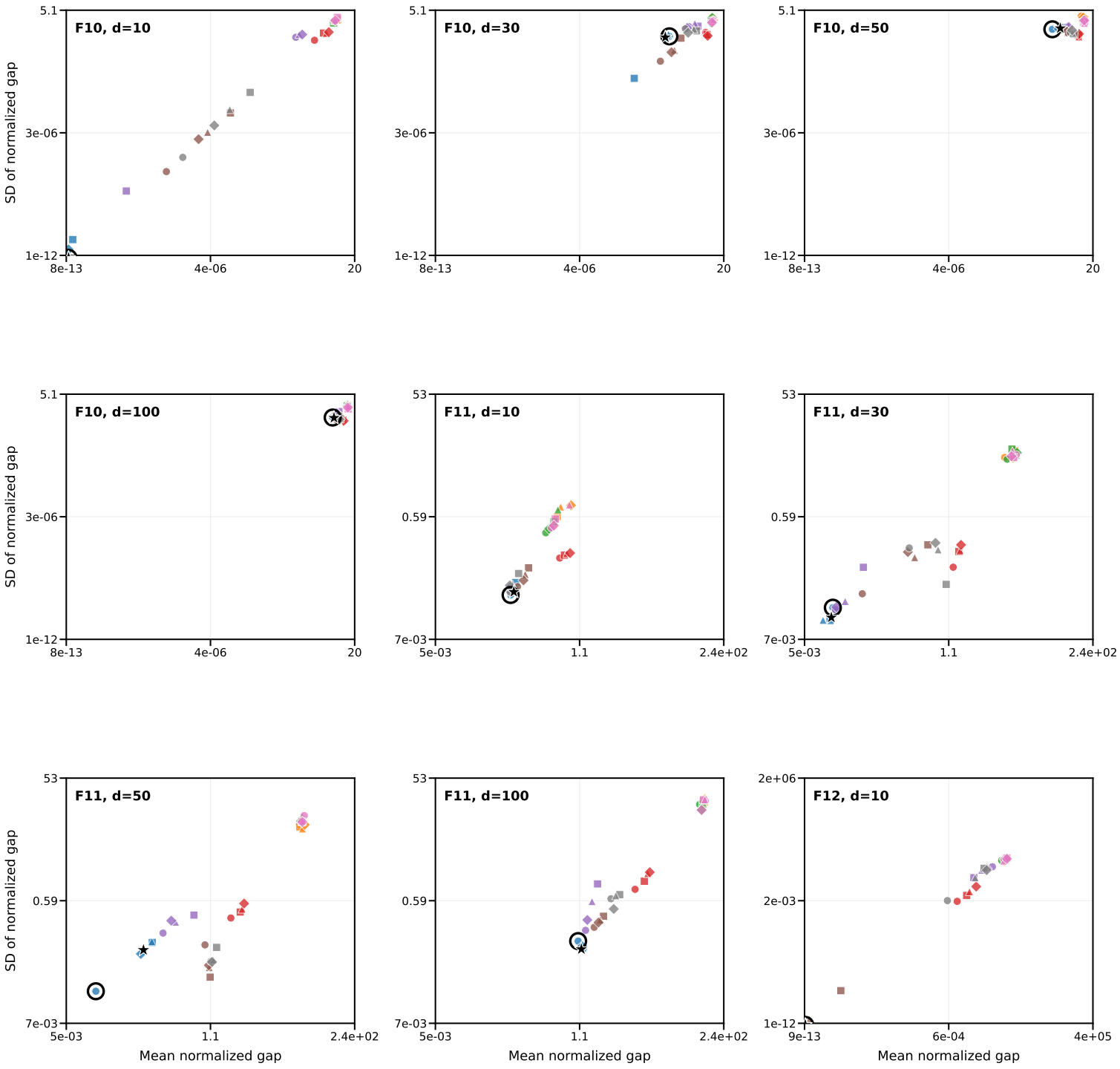
The 23-page atlas shows one page per benchmark function. Scalable functions include one panel per tested dimension with shared axis limits. Each point gives the mean and standard deviation of normalized gap over 31 runs for one configuration–instance pair. Logarithmic axes display zero at 10^{-12}. Colour identifies rule-base pattern, marker shape identifies granularity, the star marks PSO, and the open circle marks the selected FCS. Overlapping points denote equal summaries; axis ranges vary by function.

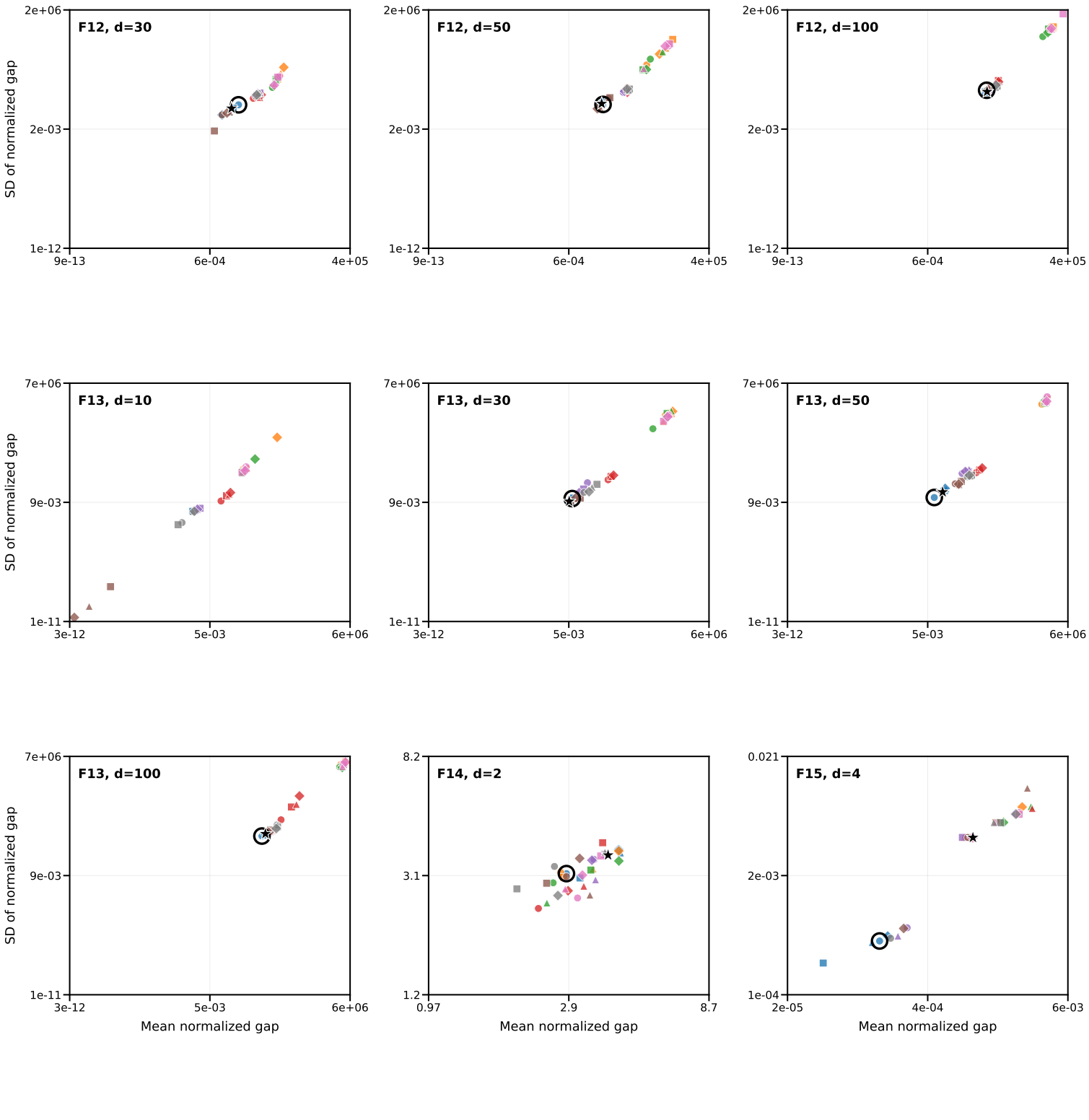

F1, d=10
F1, d=30
F1, d=50
F1, d=100
F2, d=10
F2, d=30
F2, d=50
F2, d=100
F3, d=10
SD of normalized gap
Mean normalized gap
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

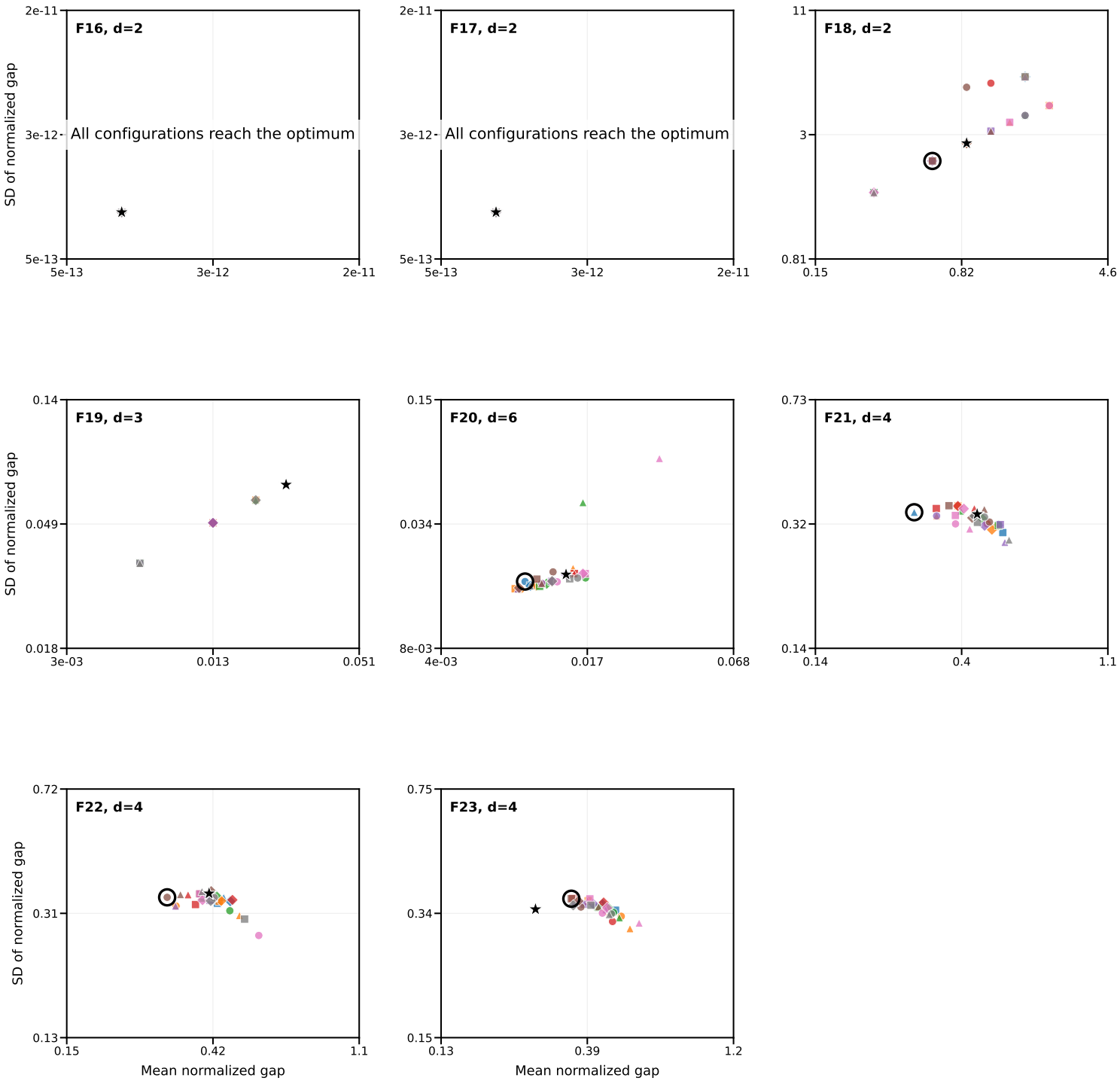

F3, d=30
F3, d=50
F3, d=100
F4, d=10
F4, d=30
F4, d=50
F4, d=100
F5, d=10
F5, d=30
SD of normalized gap
Mean normalized gap
8e+04
6e-04
5e-12
2e-12
4e-04
7e+04
21
1e-04
9e-10
4e-10
42
4e+06
2e+03
0.66
2.2
4e+03
9e+06
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function


F5, d=50
F5, d=100
F6, d=10
F6, d=30
F6, d=50
F6, d=100
F7, d=10
F7, d=30
F7, d=50
SD of normalized gap
Mean normalized gap
4e+06
2e+03
0.66
2.2
4e+03
9e+06
6e+03
6e-05
7e-13
4e-12
3e-04
3e+04
0.19
8e-03
3e-04
6e-04
0.024
0.86
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function


F7, d=100
F8, d=10
F8, d=30
F8, d=50
F8, d=100
F9, d=10
F9, d=30
F9, d=50
F9, d=100
SD of normalized gap
Mean normalized gap
0.19
8e-03
3e-04
6e-04
0.024
0.86
0.2
0.062
0.019
0.49
1.2
2.4e+02
22
1.9
4.8
72
1e+03
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

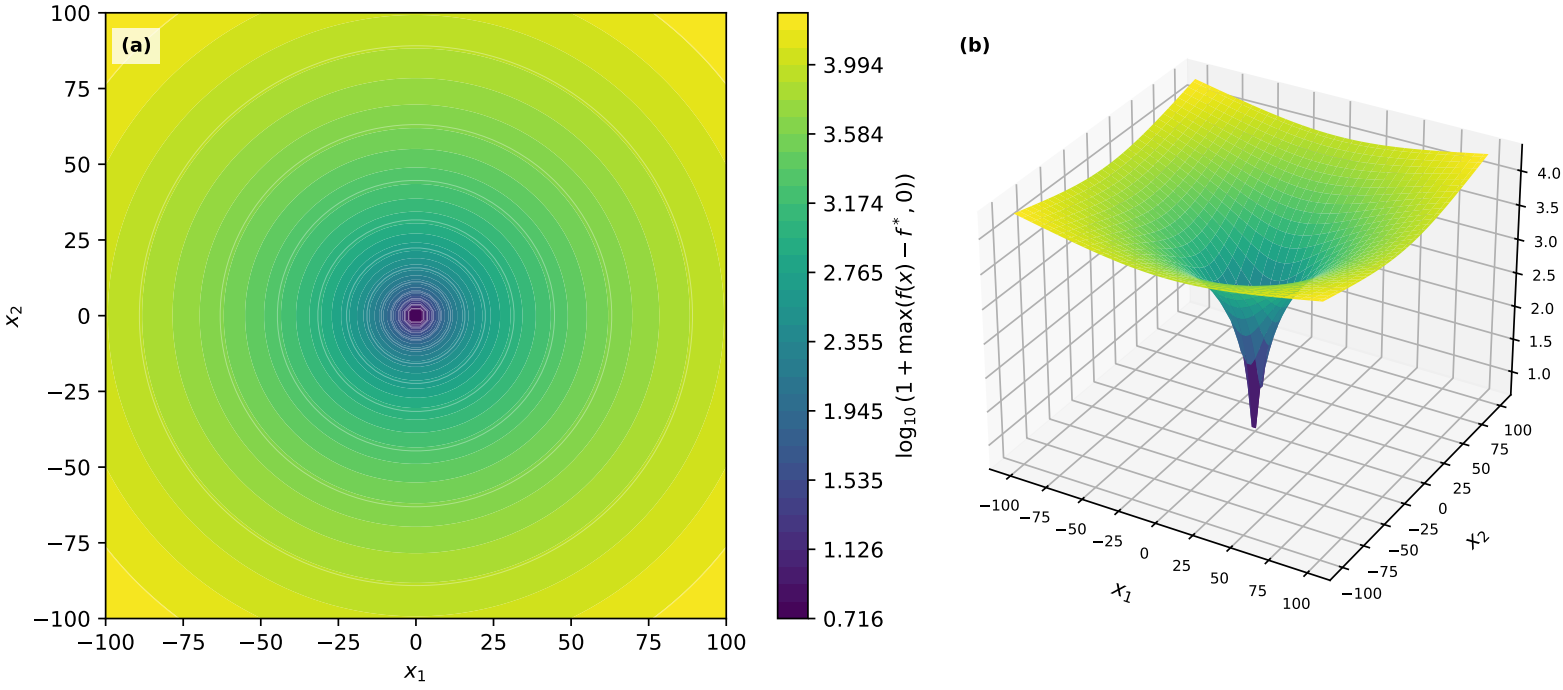

F10, d=10
F10, d=30
F10, d=50
F10, d=100
F11, d=10
F11, d=30
F11, d=50
F11, d=100
F12, d=10
SD of normalized gap
Mean normalized gap
5.1
3e-06
1e-12
8e-13
4e-06
20
53
0.59
7e-03
5e-03
1.1
2.4e+02
2e+06
2e-03
9e-13
6e-04
4e+05
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

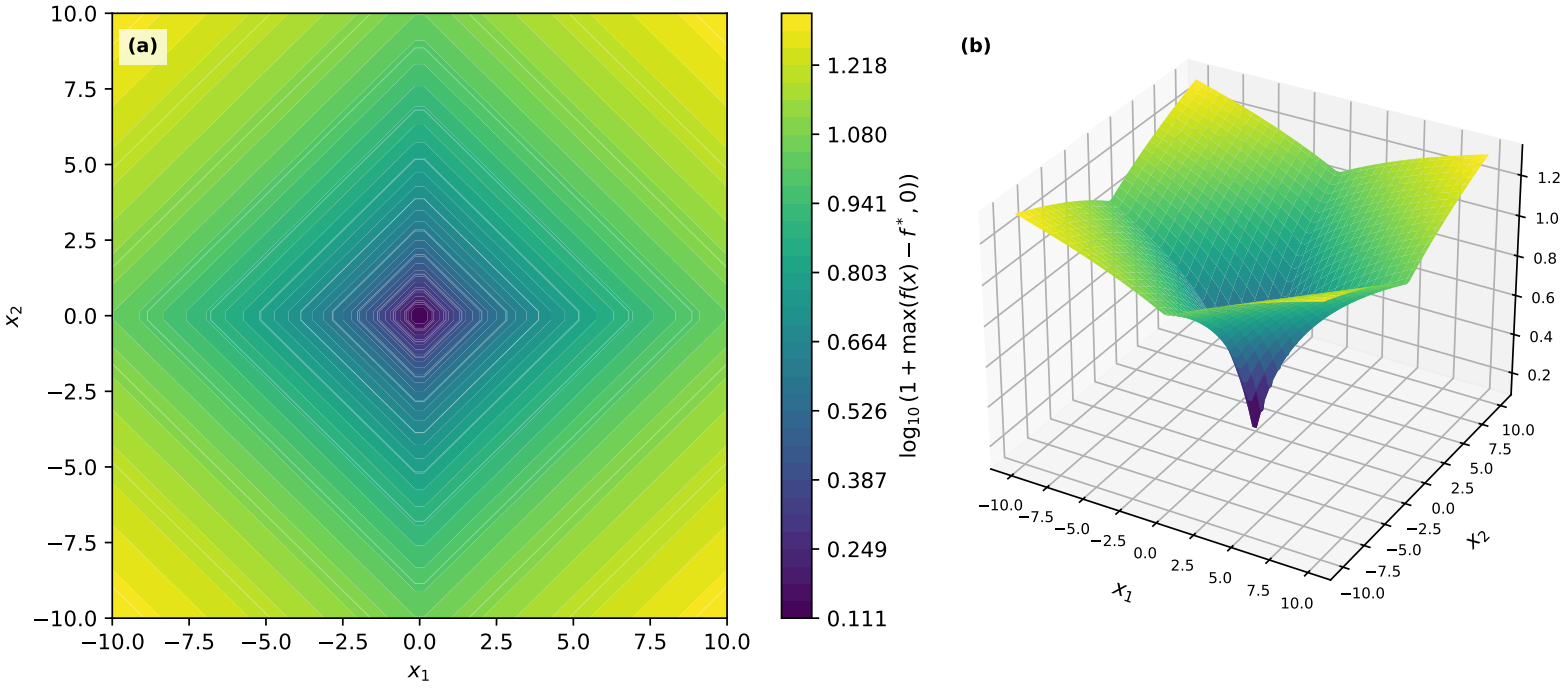

F12, d=30
F12, d=50
F12, d=100
F13, d=10
F13, d=30
F13, d=50
F13, d=100
F14, d=2
F15, d=4
SD of normalized gap
Mean normalized gap
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

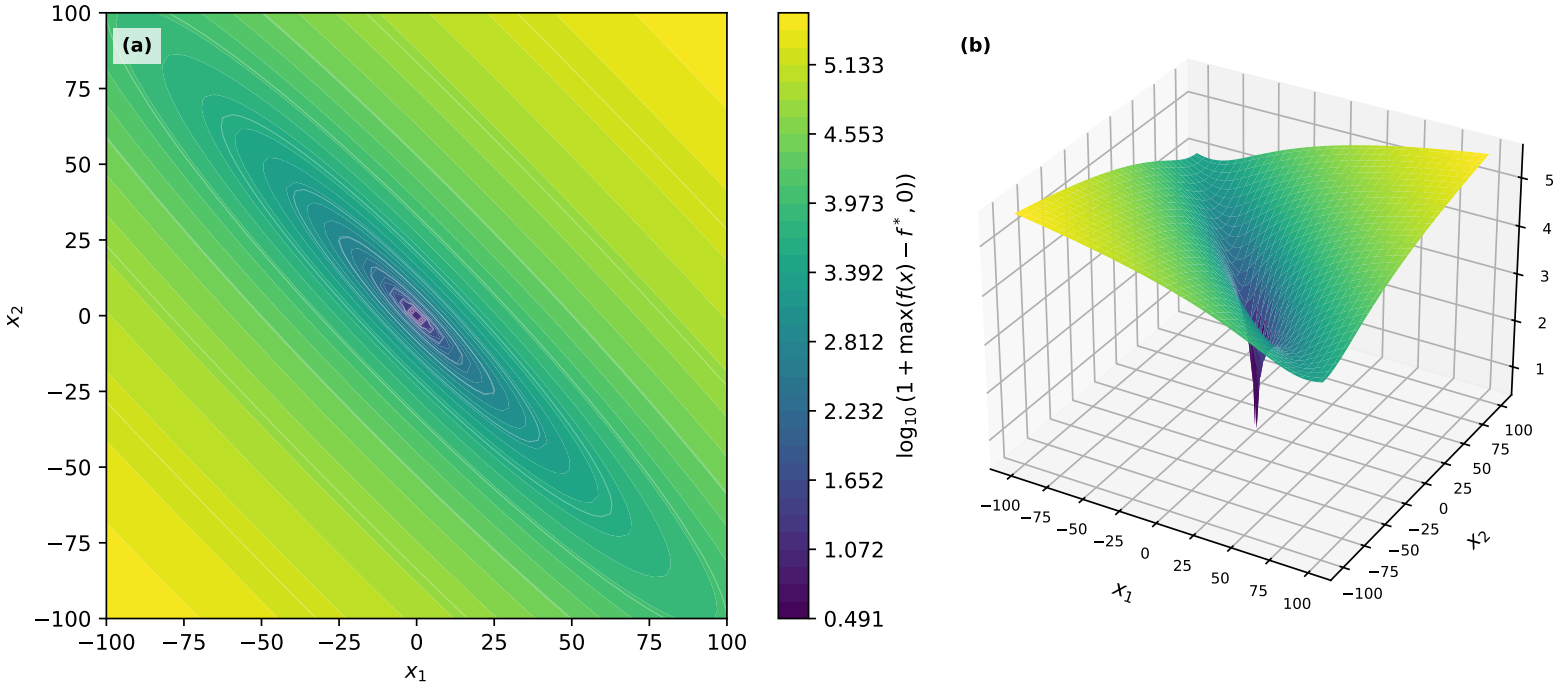

F16, d=2
All configurations reach the optimum
F17, d=2
All configurations reach the optimum
F18, d=2
F19, d=3
F20, d=6
F21, d=4
F22, d=4
F23, d=4
SD of normalized gap
Mean normalized gap
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

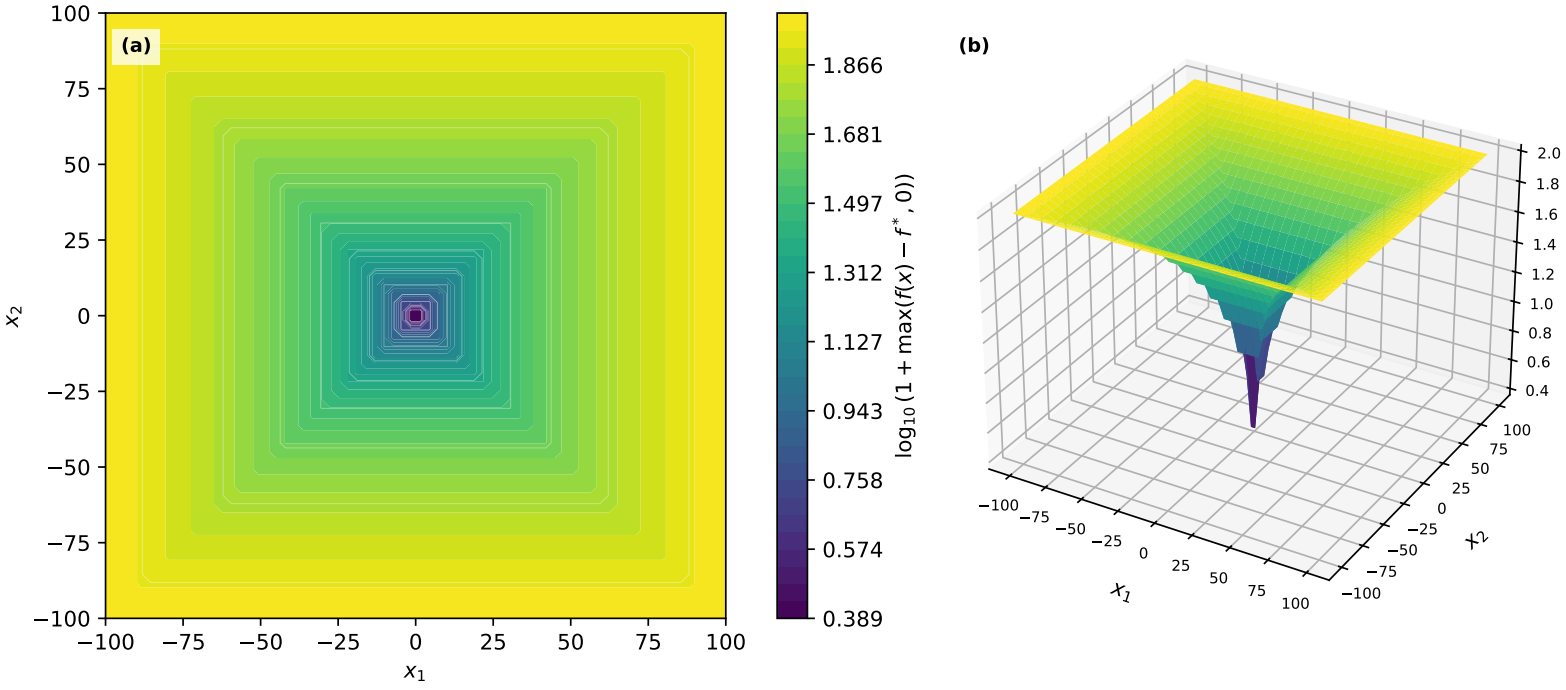
4 Benchmark-function landscape slices

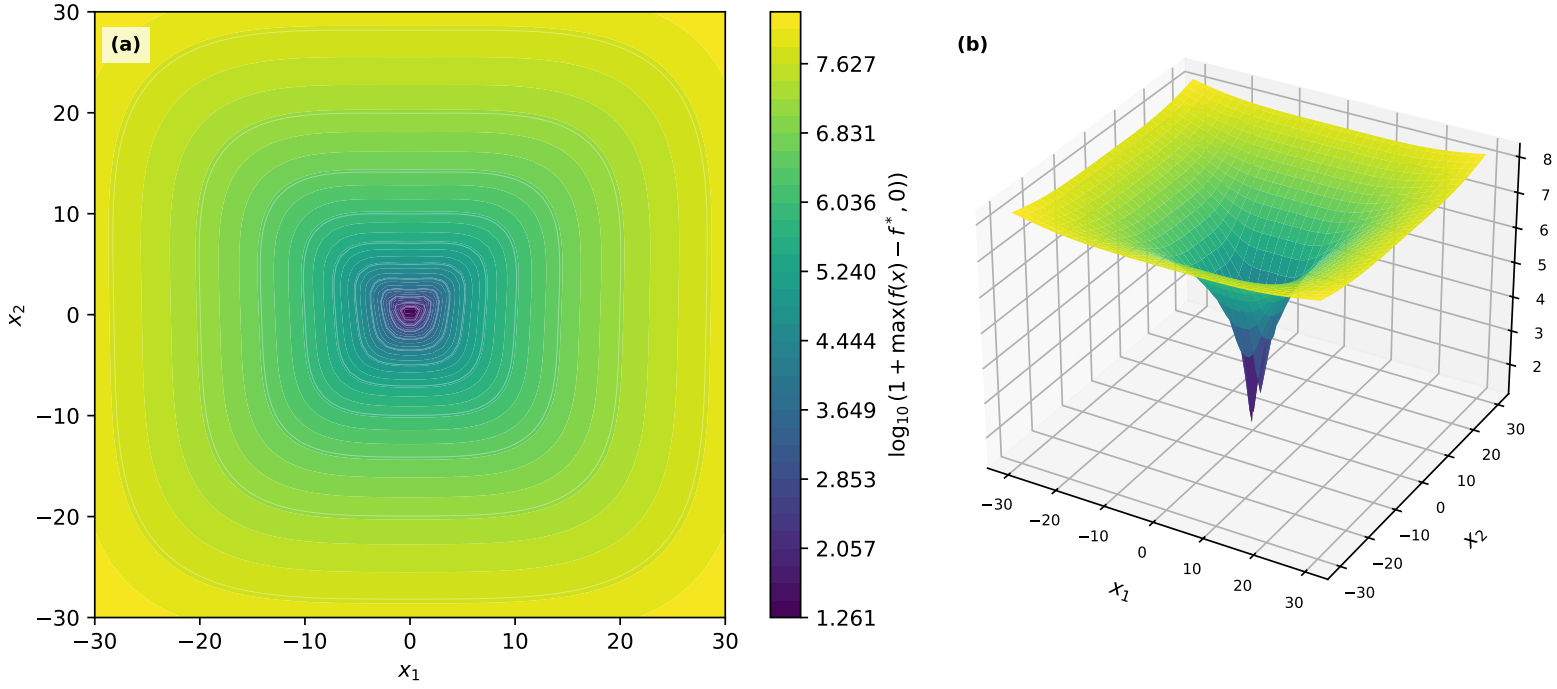
Each page pairs a contour with a 3-D view of the same two-dimensional slice. Coordinates use the benchmark bounds; dimensions beyond two are fixed at the bounds midpoint. Elevation and colour show $\log_{10}(1 + \max(f(x) - f^*, 0))$. F8 uses its dimension-scaled optimum, and F7 shows the expected objective with uniform-noise mean 0.5.

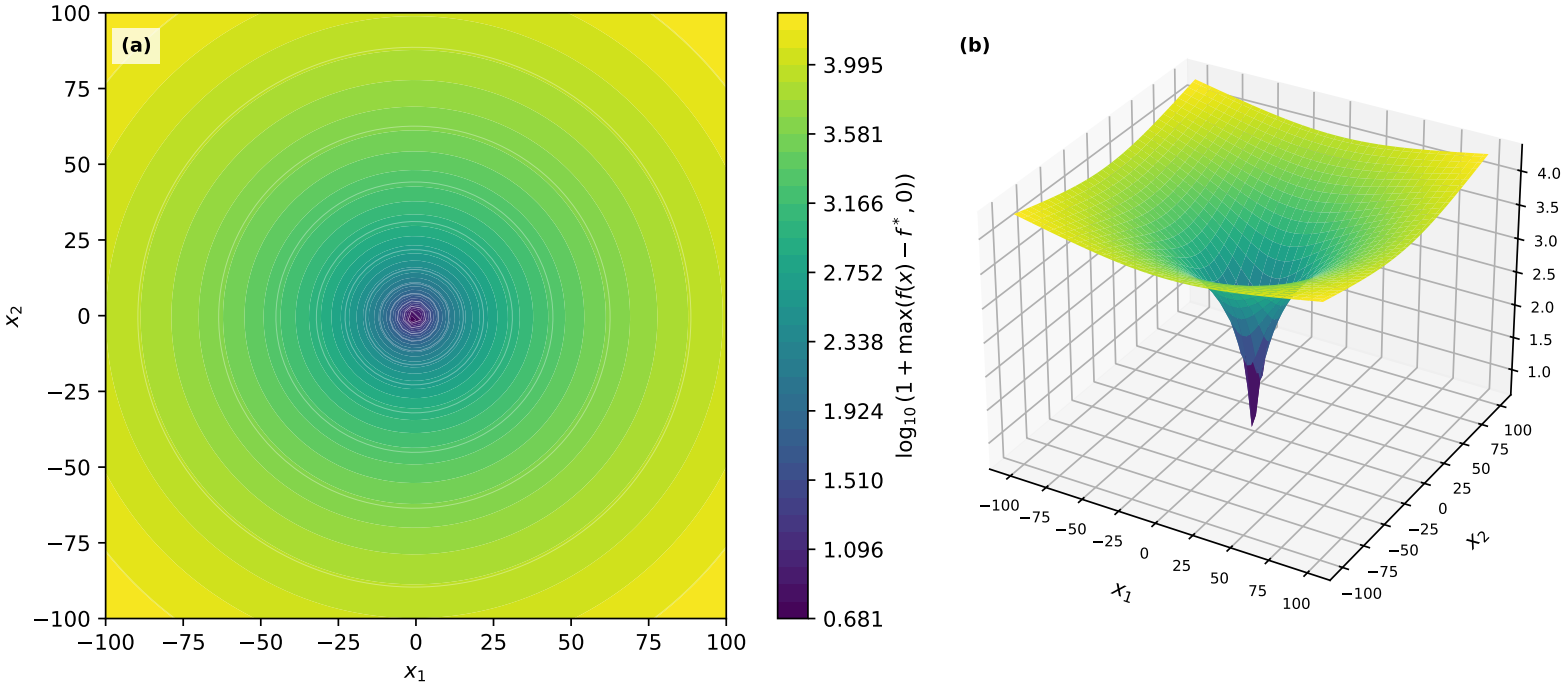

(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
3.994
3.584
3.174
2.765
2.355
1.945
1.535
1.126
0.716

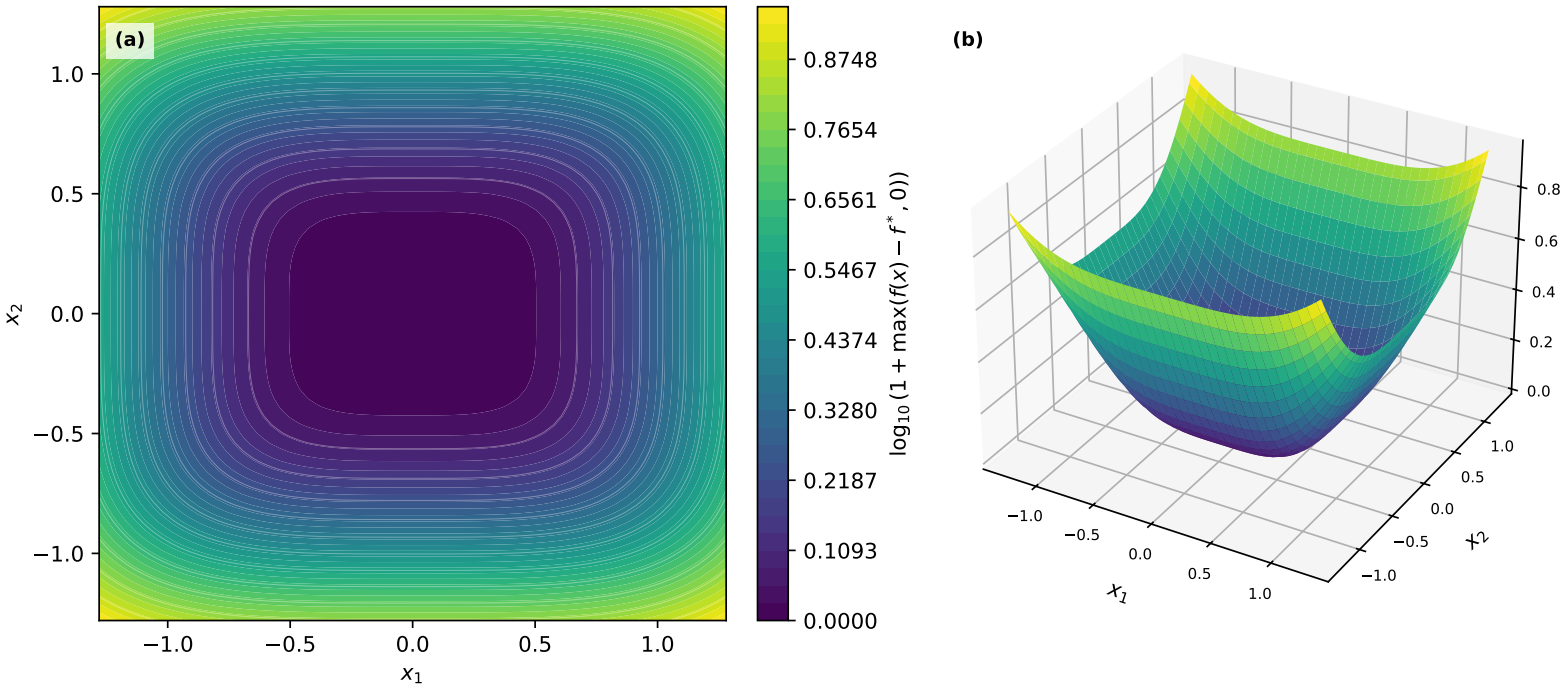

(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
1.218
1.080
0.941
0.803
0.664
0.526
0.387
0.249
0.111

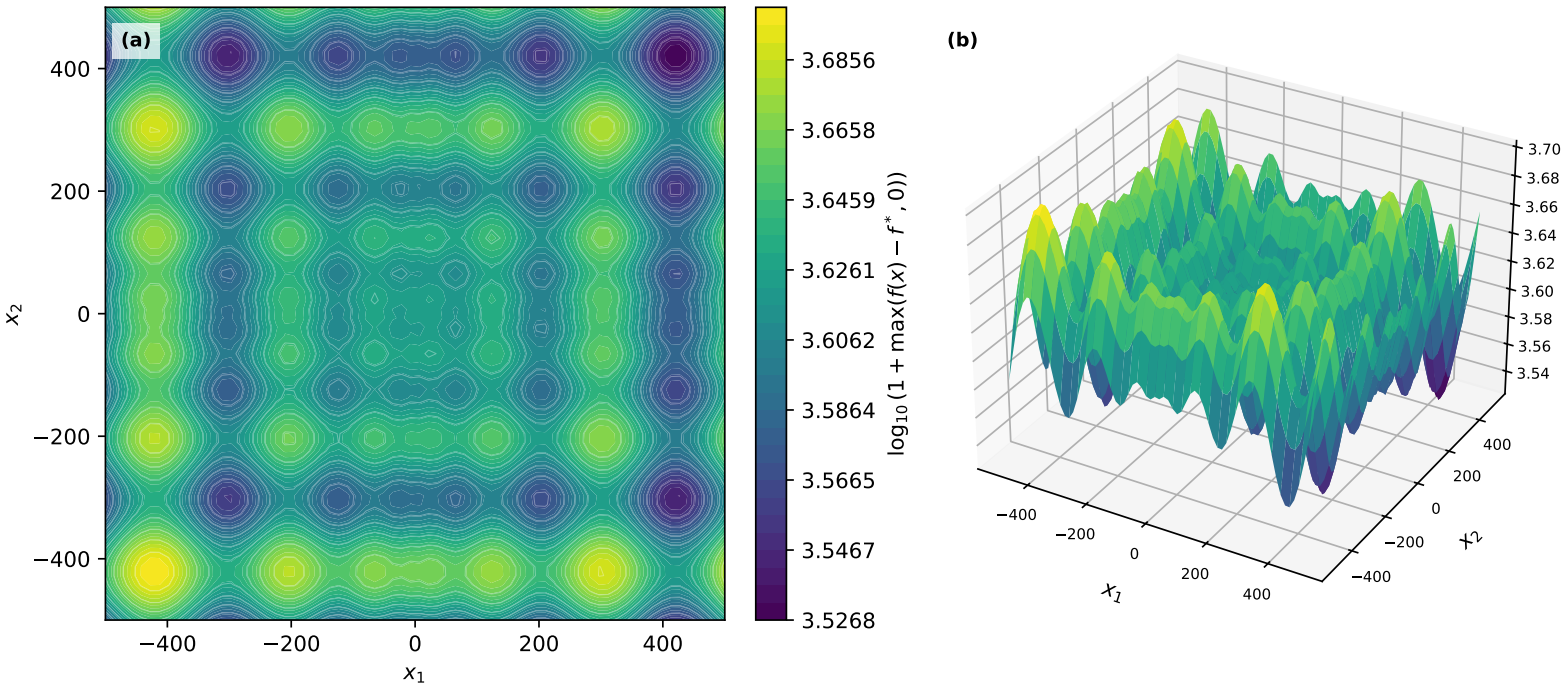

(a)
(b)
x_1
x_2
$\log_{10}(1+\max(f(x)-f^*,0))$
5.133
4.553
3.973
3.392
2.812
2.232
1.652
1.072
0.491

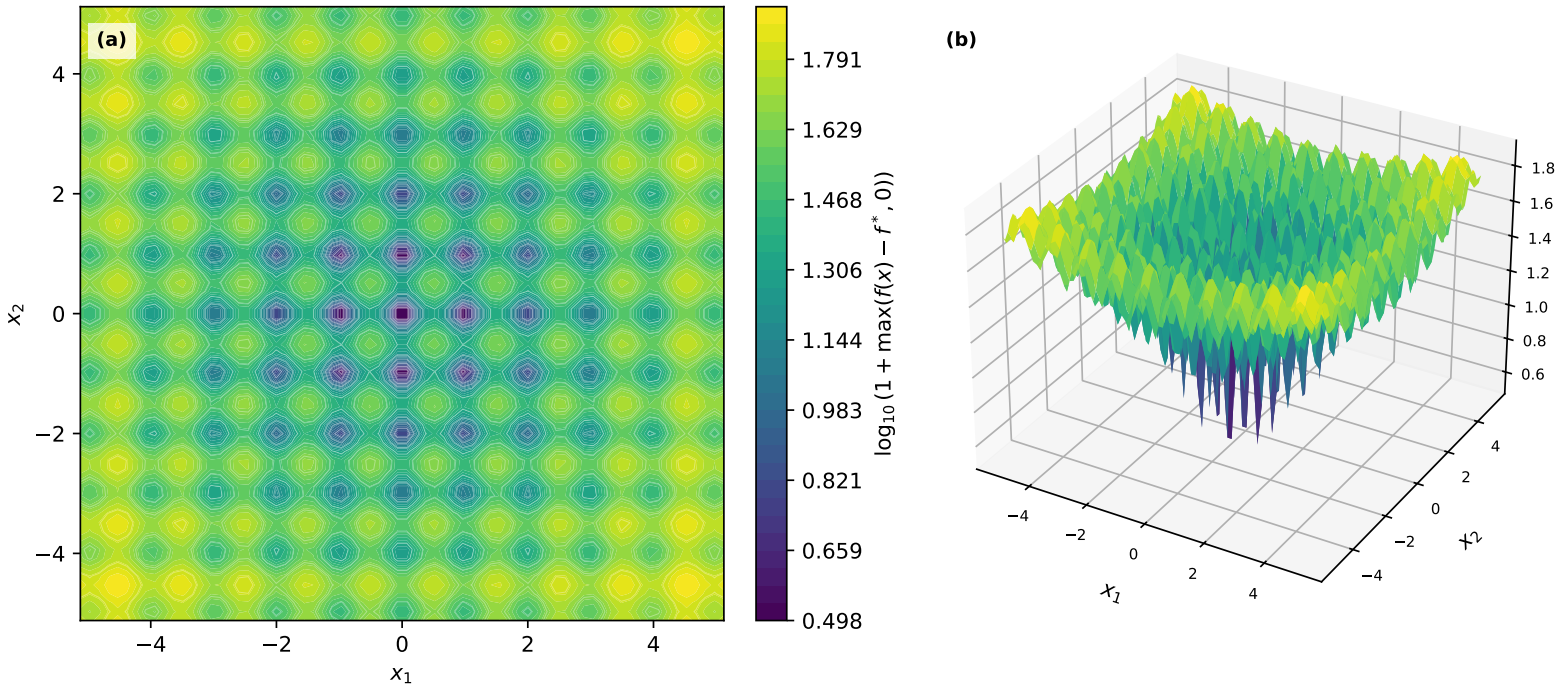

(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
1.866
1.681
1.497
1.312
1.127
0.943
0.758
0.574
0.389

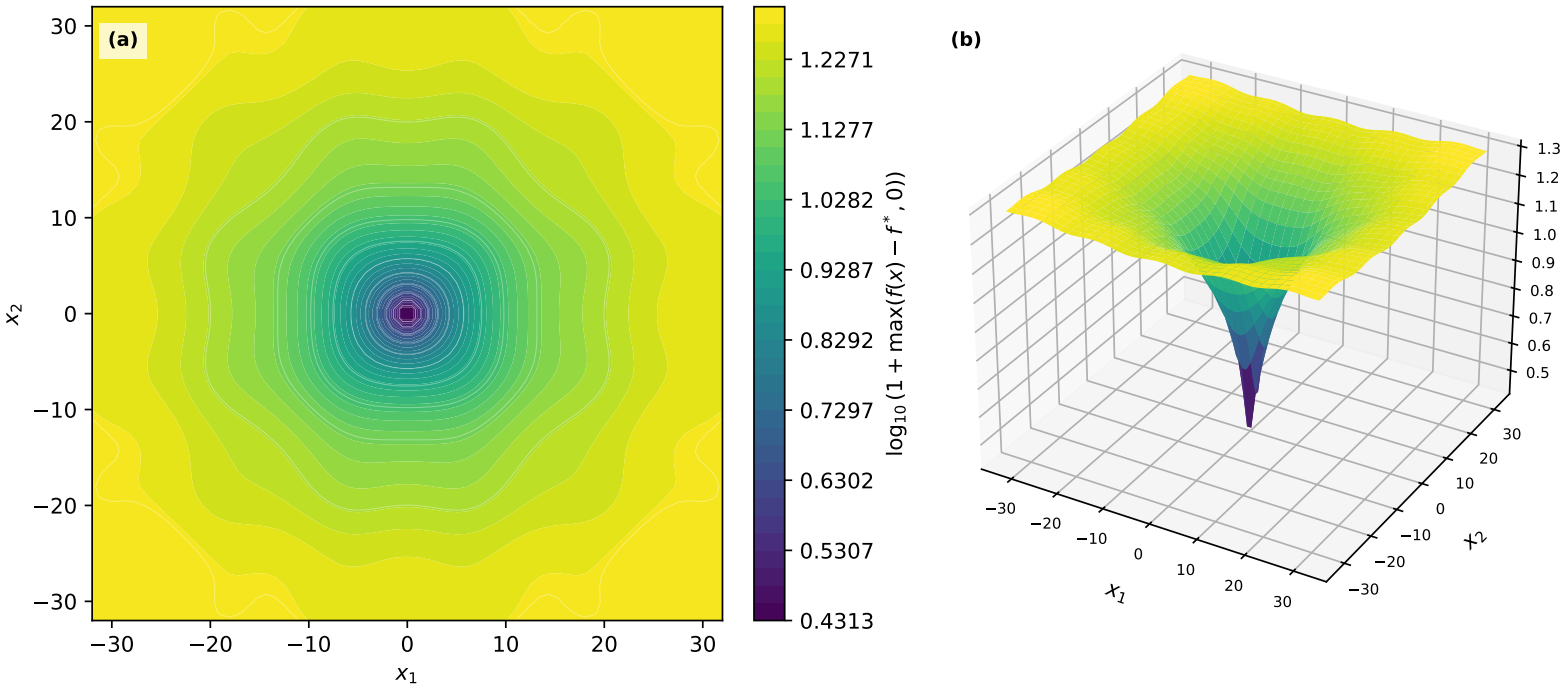

(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
7.627
6.831
6.036
5.240
4.444
3.649
2.853
2.057
1.261

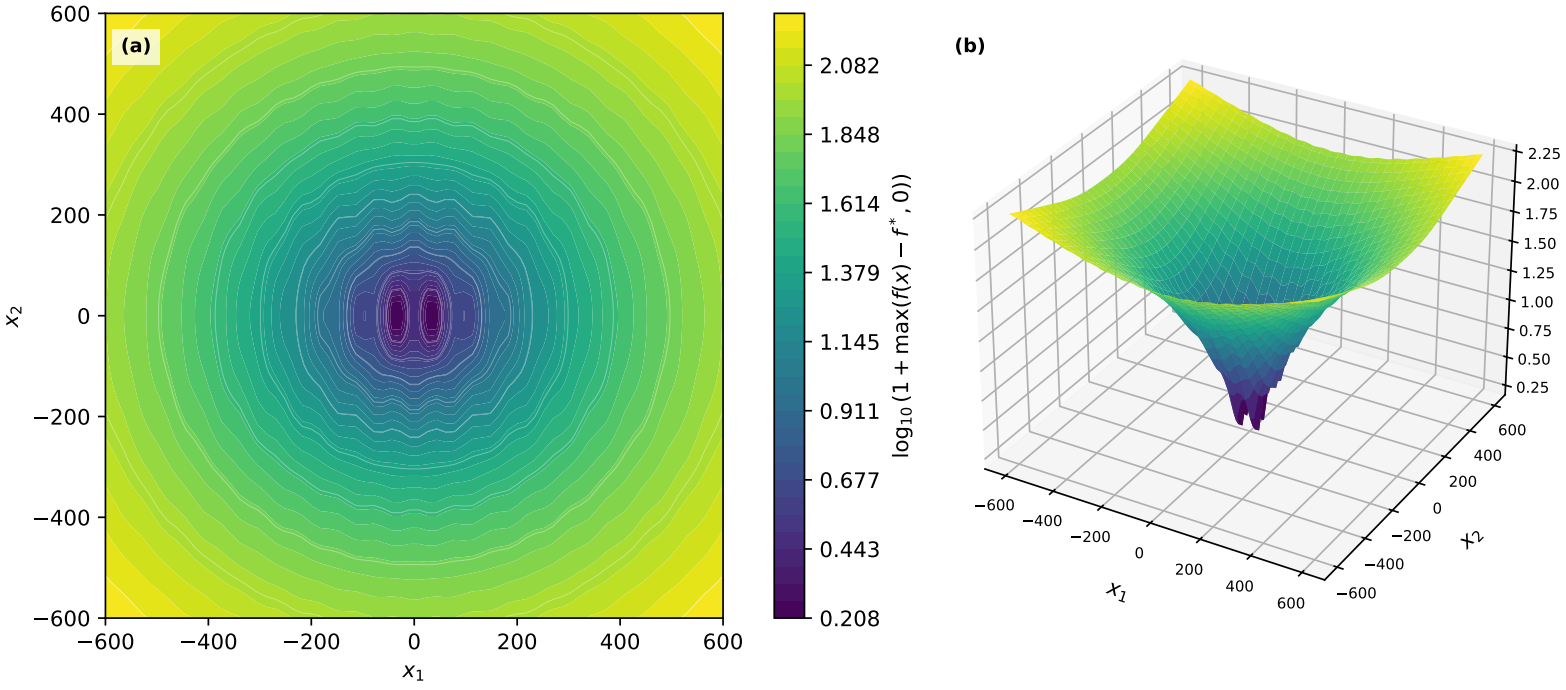

(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
3.995
3.581
3.166
2.752
2.338
1.924
1.510
1.096
0.681

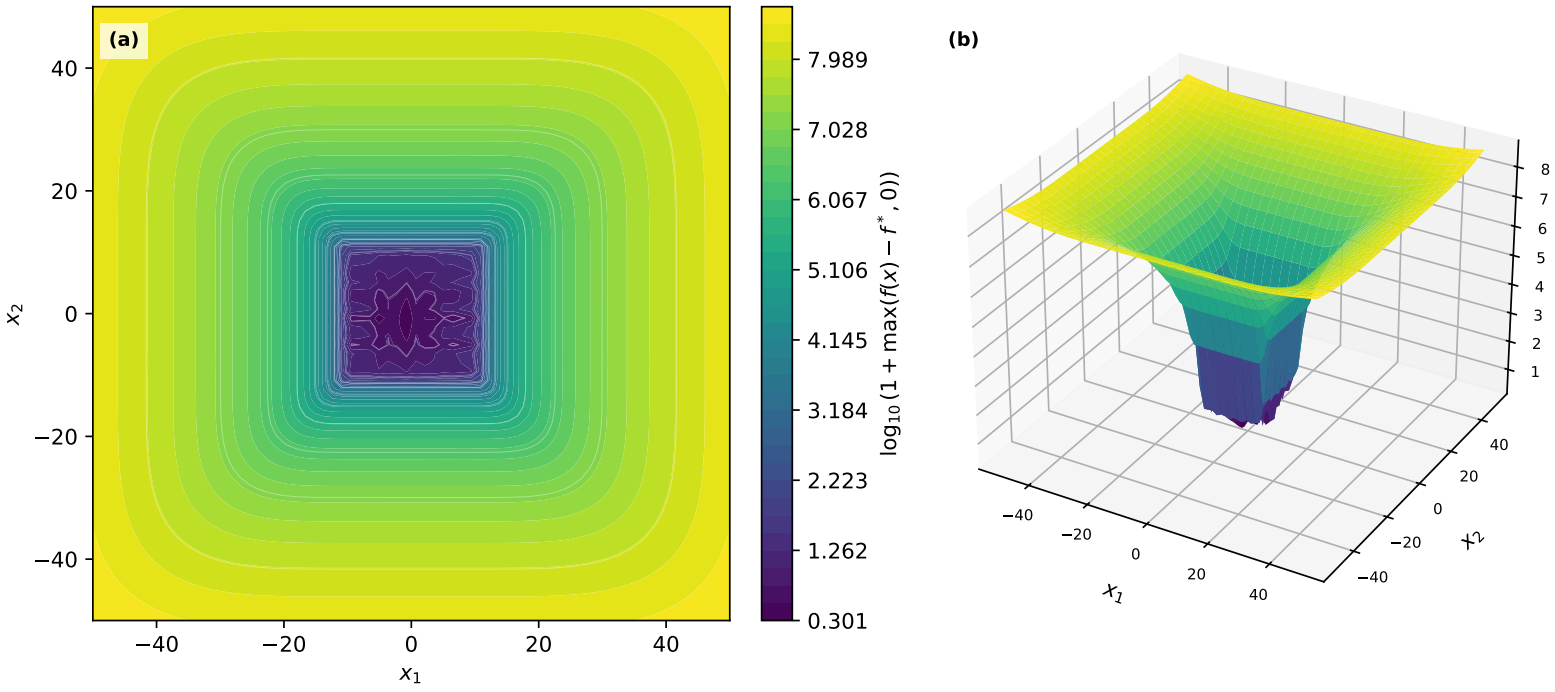

(a)
1.0
0.5
0.0
−0.5
−1.0
x_2
−1.0
−0.5
0.0
0.5
1.0
x_1
0.8748
0.7654
0.6561
0.5467
0.4374
0.3280
0.2187
0.1093
0.0000
$\log_{10}(1 + \max(f(x) - f^*, 0))$
(b)
−1.0
−0.5
0.0
0.5
1.0
x_1
−1.0
−0.5
0.0
0.5
1.0
x_2
0.0
0.2
0.4
0.6
0.8

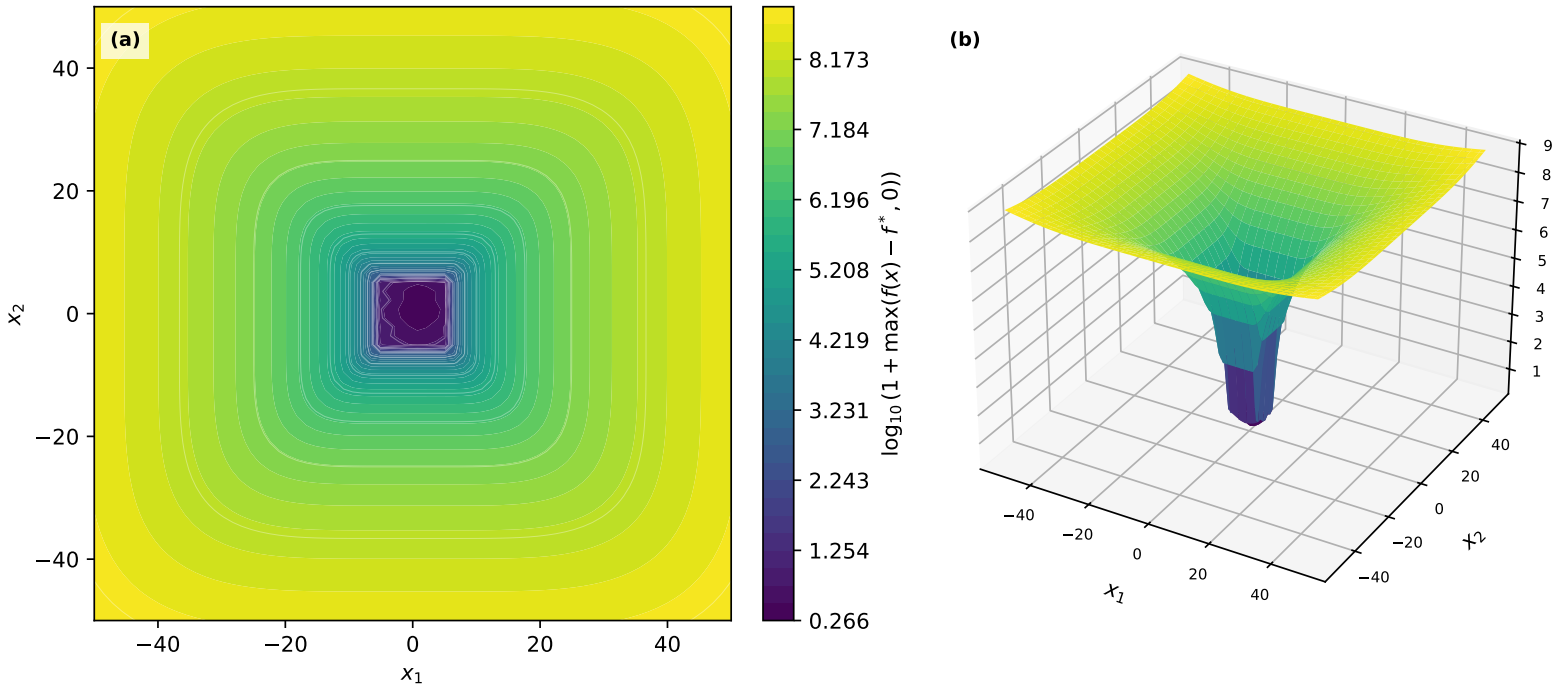

(a)
400
200
0
−200
−400
x_2
−400
−200
0
200
400
x_1
3.6856
3.6658
3.6459
3.6261
3.6062
3.5864
3.5665
3.5467
3.5268
$\log_{10}(1 + \max(f(x) - f^*, 0))$
(b)
3.70
3.68
3.66
3.64
3.62
3.60
3.58
3.56
3.54
−400
−200
0
200
400
x_1
400
200
0
−200
−400
x_2

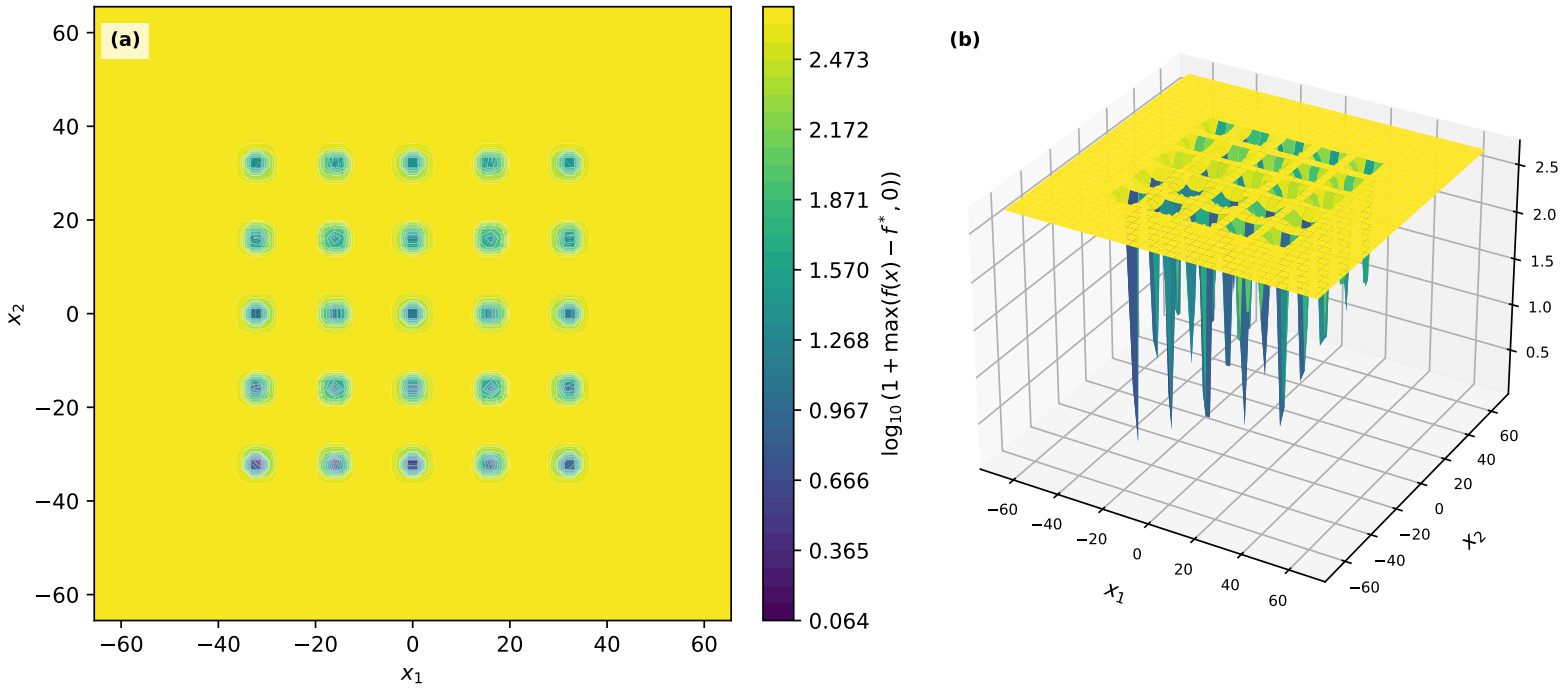

(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
1.791
1.629
1.468
1.306
1.144
0.983
0.821
0.659
0.498

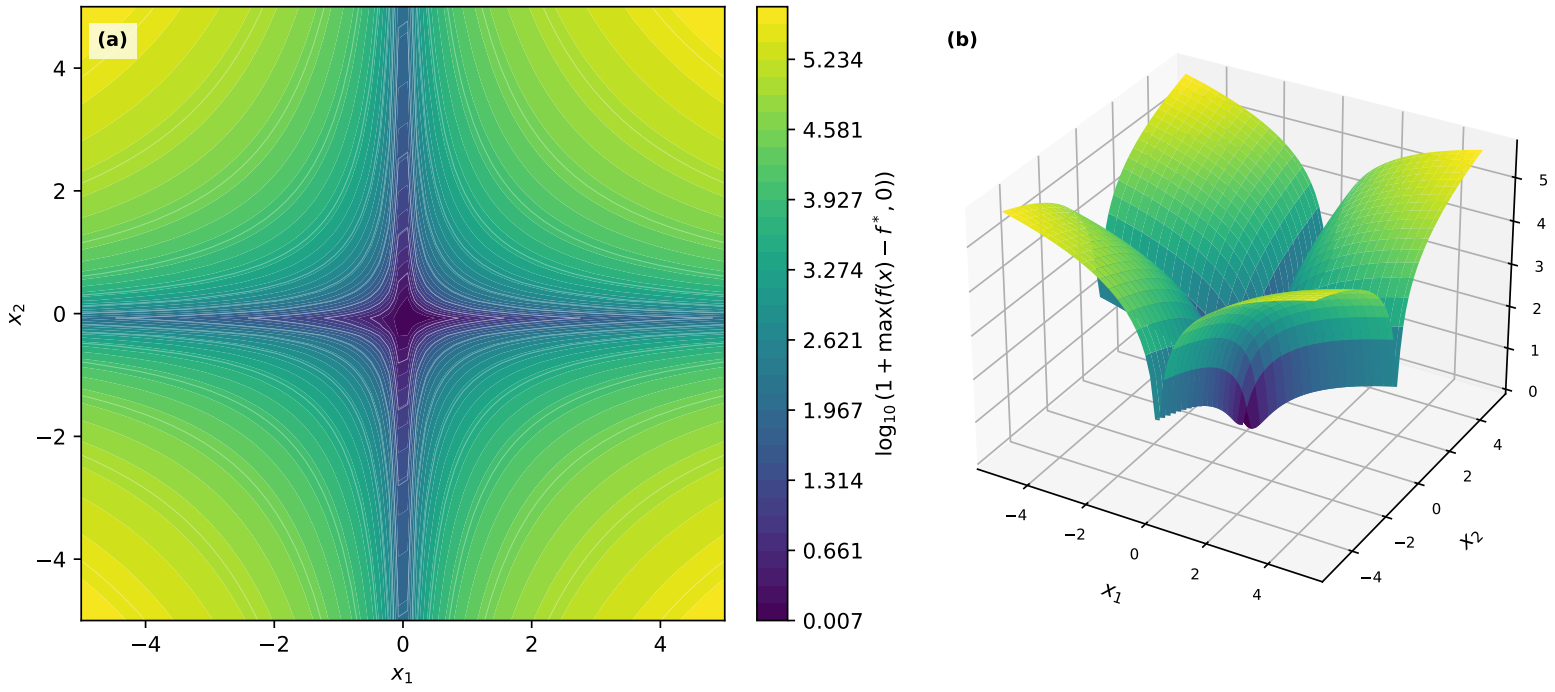

(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
1.2271
1.1277
1.0282
0.9287
0.8292
0.7297
0.6302
0.5307
0.4313

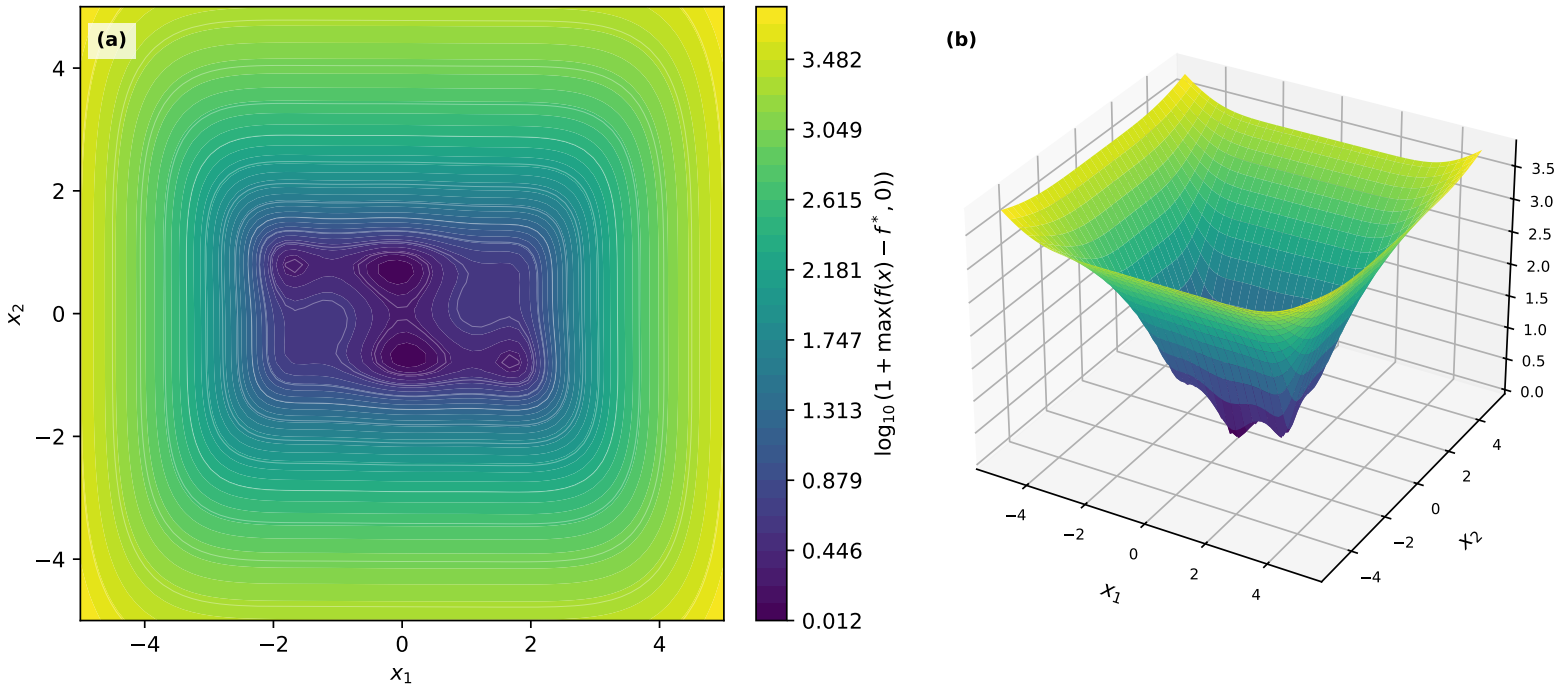

(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
2.082
1.848
1.614
1.379
1.145
0.911
0.677
0.443
0.208
600
400
200
0
−200
−400
−600
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25

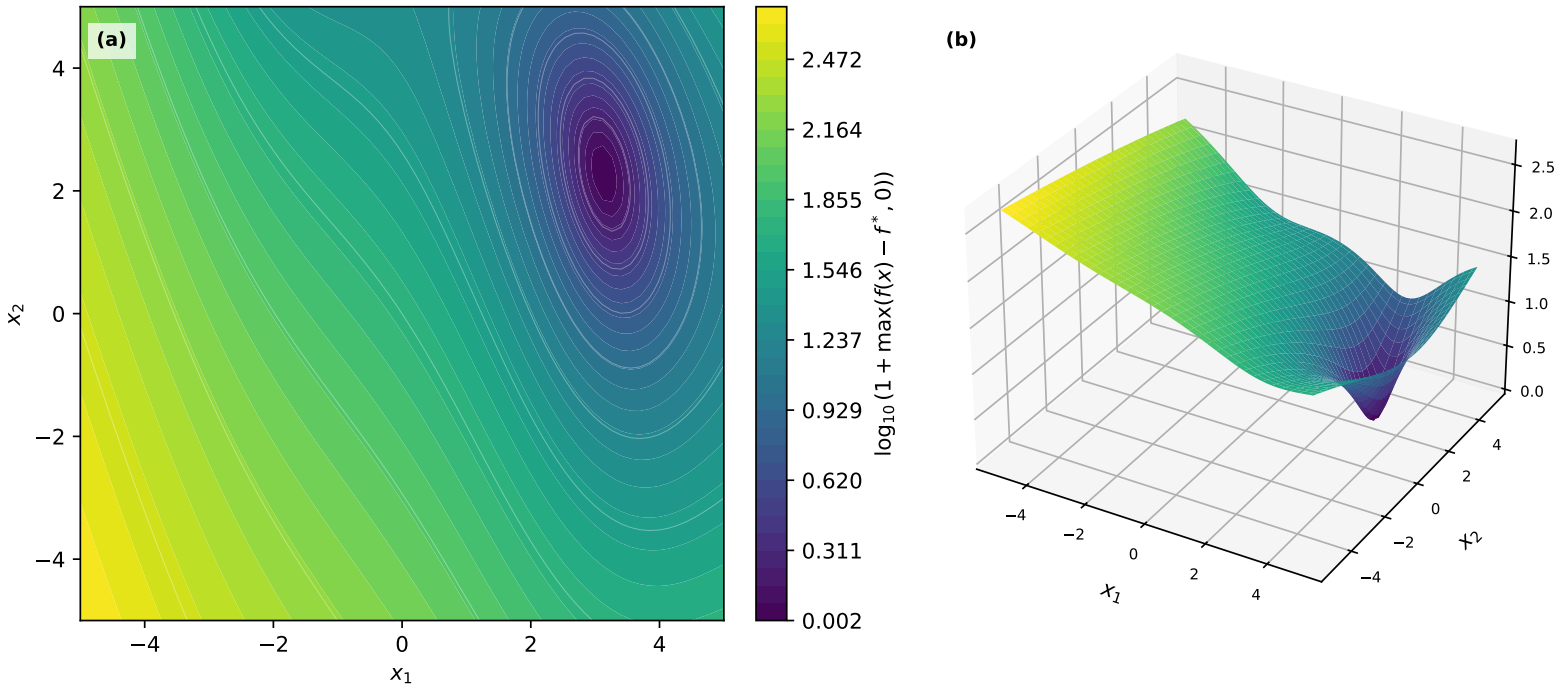

(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
7.989
7.028
6.067
5.106
4.145
3.184
2.223
1.262
0.301

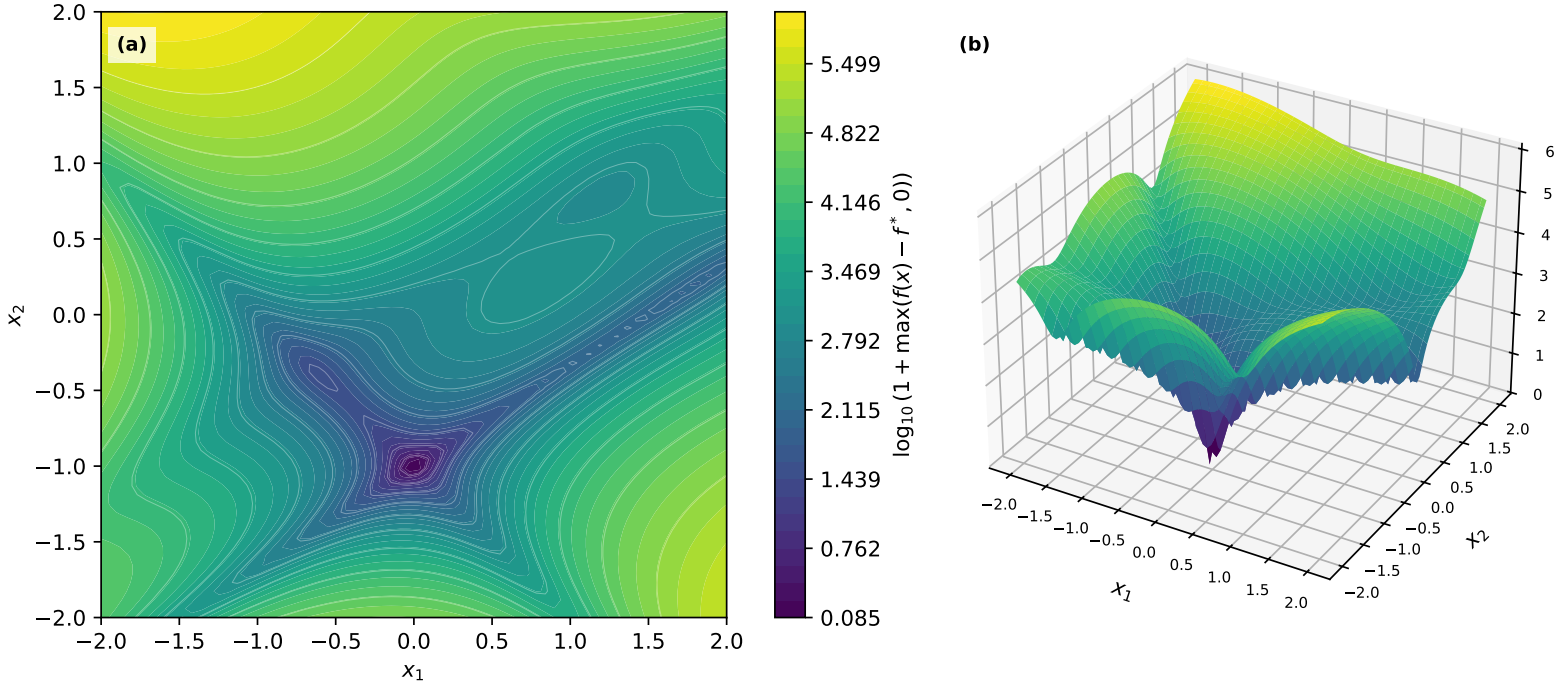

(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
8.173
7.184
6.196
5.208
4.219
3.231
2.243
1.254
0.266

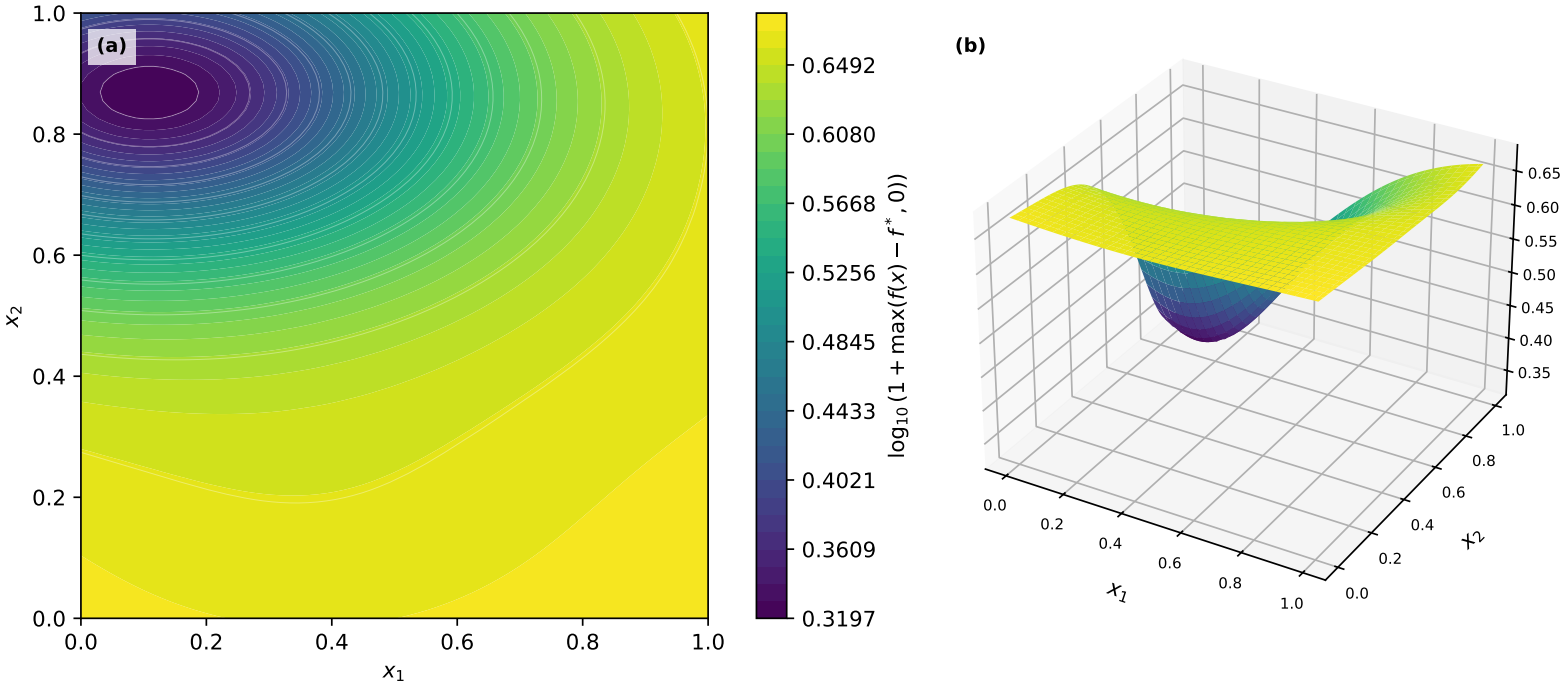

(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
2.473
2.172
1.871
1.570
1.268
0.967
0.666
0.365
0.064

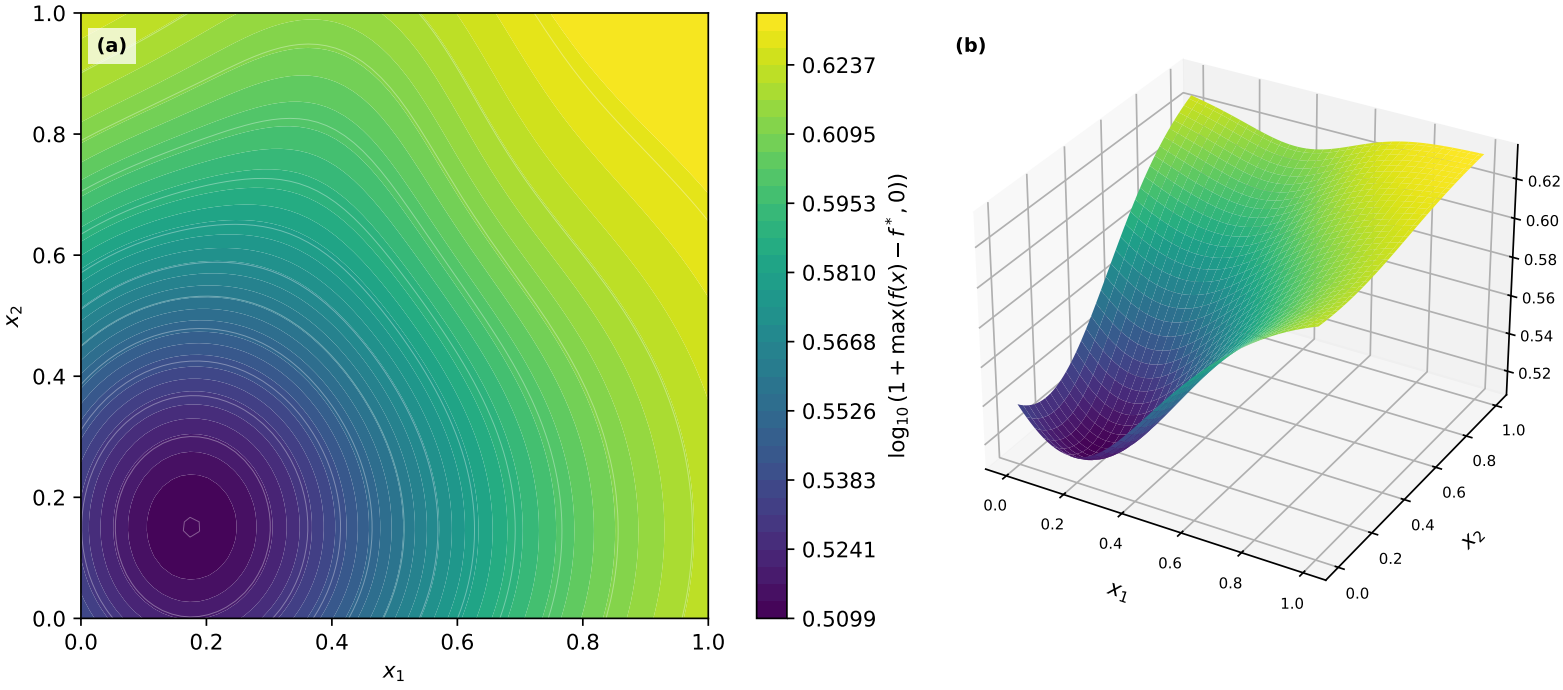

(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
5.234
4.581
3.927
3.274
2.621
1.967
1.314
0.661
0.007

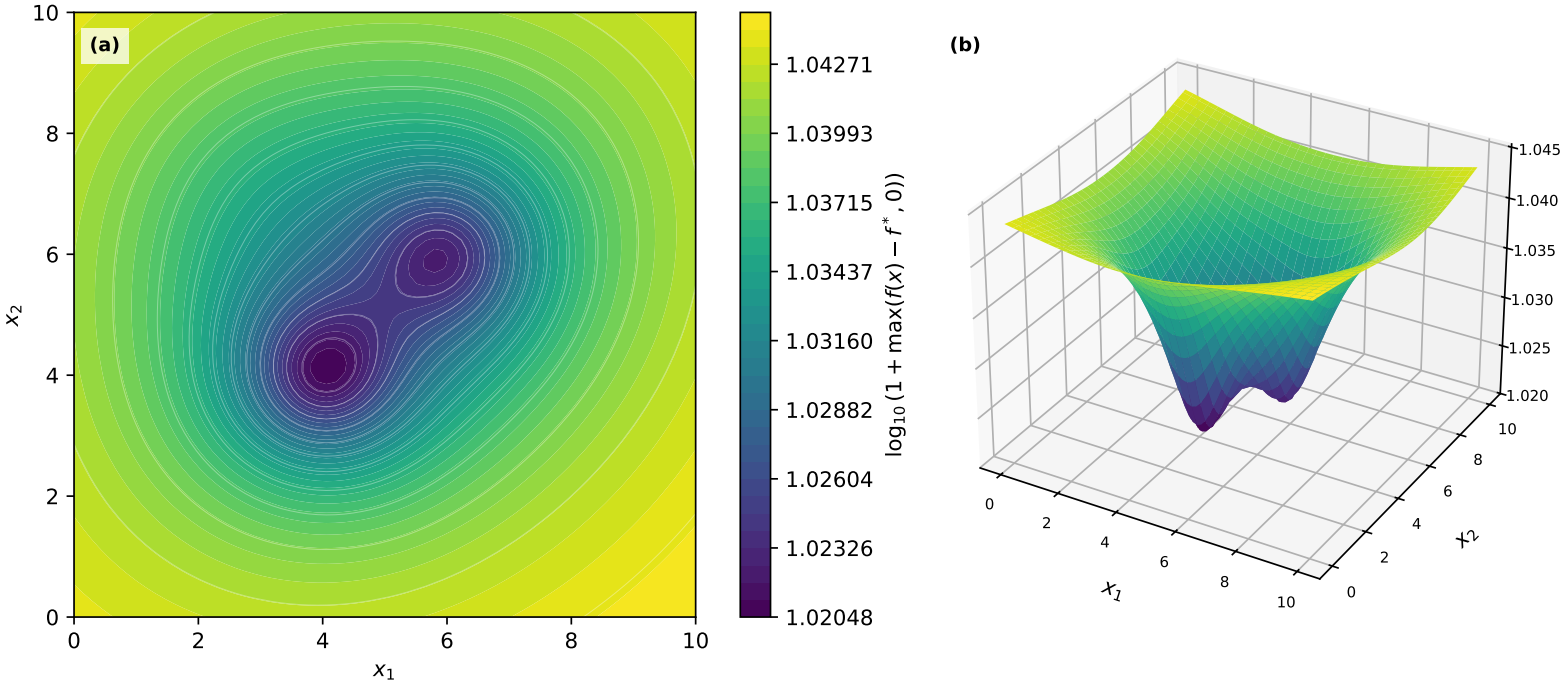

(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
3.482
3.049
2.615
2.181
1.747
1.313
0.879
0.446
0.012

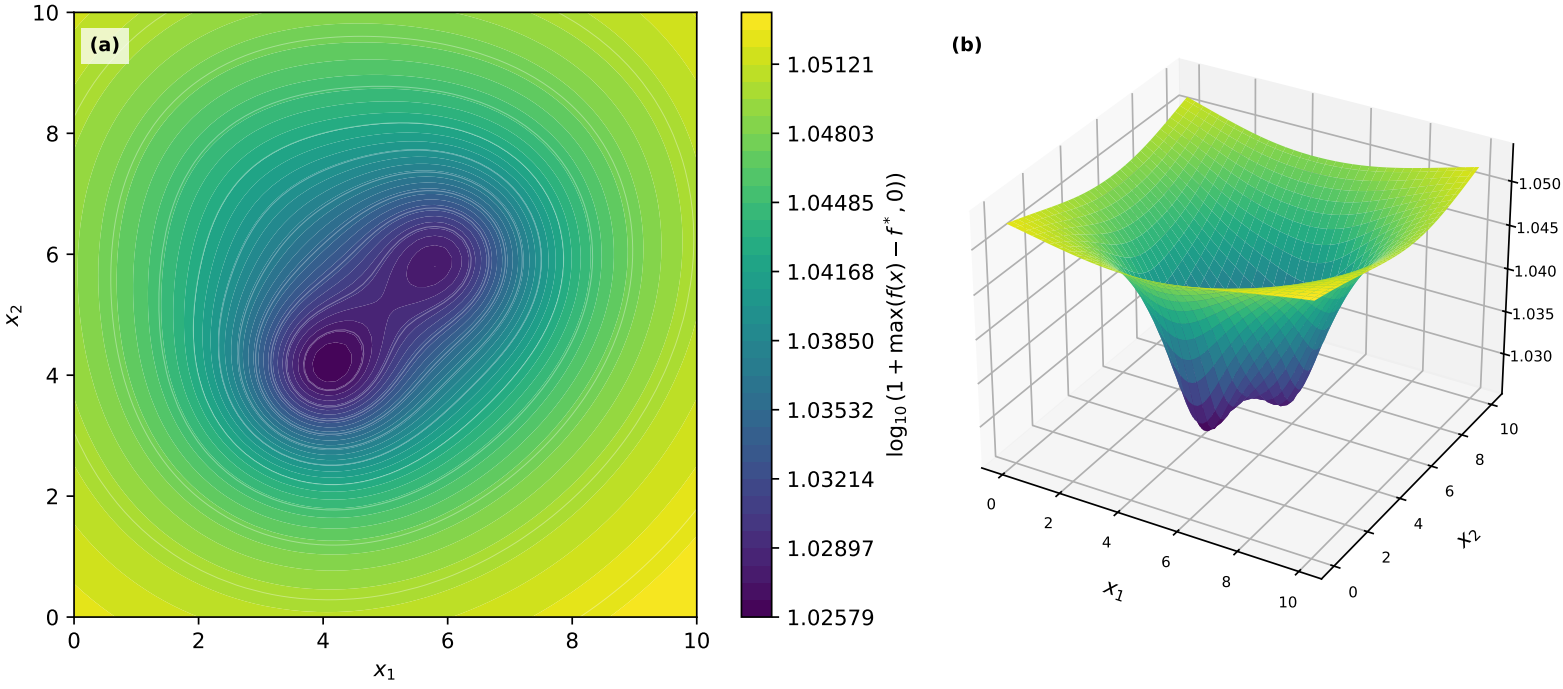

(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
2.472
2.164
1.855
1.546
1.237
0.929
0.620
0.311
0.002

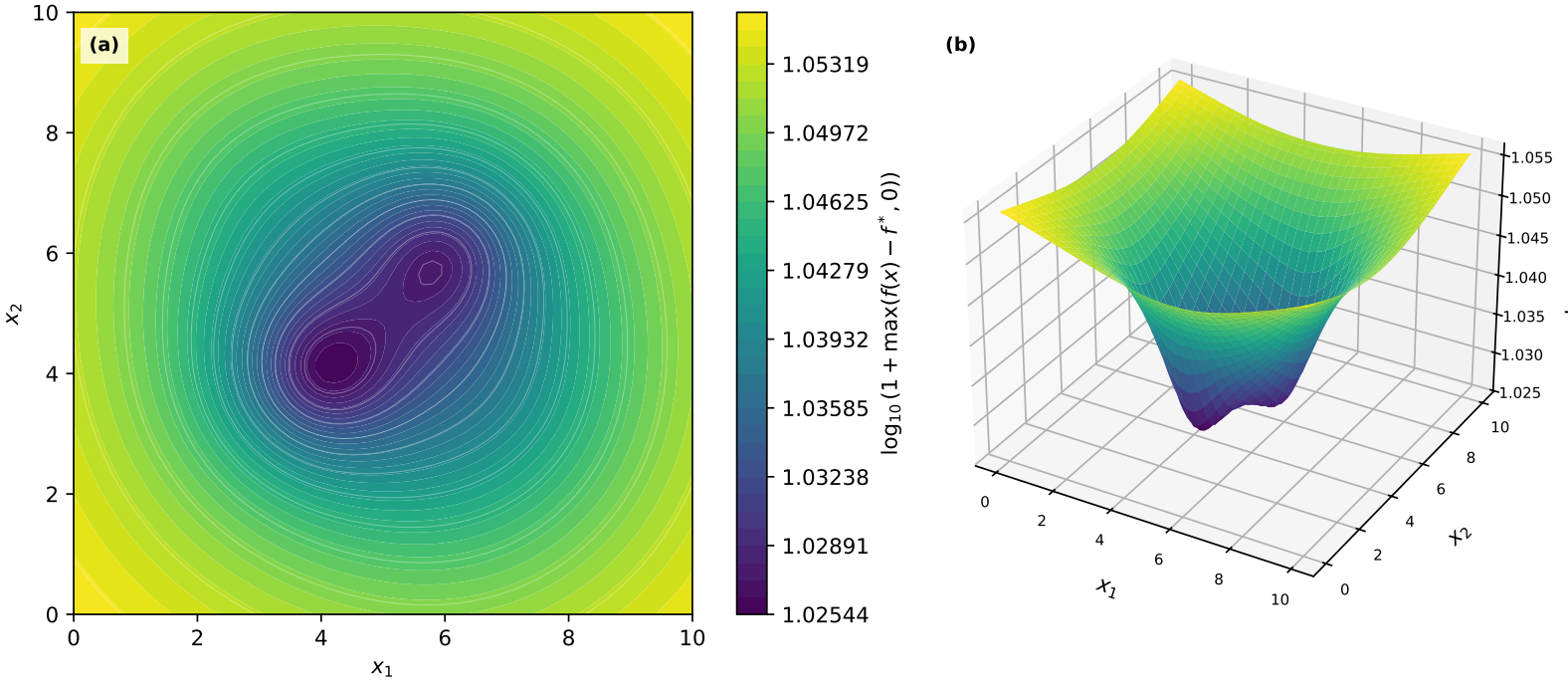

(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
5.499
4.822
4.146
3.469
2.792
2.115
1.439
0.762
0.085


(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
0.6492
0.6080
0.5668
0.5256
0.4845
0.4433
0.4021
0.3609
0.3197


(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
0.6237
0.6095
0.5953
0.5810
0.5668
0.5526
0.5383
0.5241
0.5099


(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
1.04271
1.03993
1.03715
1.03437
1.03160
1.02882
1.02604
1.02326
1.02048
1.045
1.040
1.035
1.030
1.025
1.020


(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
1.05121
1.04803
1.04485
1.04168
1.03850
1.03532
1.03214
1.02897
1.02579
1.050
1.045
1.040
1.035
1.030


(a)
(b)
x_1
x_2
$\log_{10}(1 + \max(f(x) - f^*, 0))$
1.05319
1.04972
1.04625
1.04279
1.03932
1.03585
1.03238
1.02891
1.02544
1.055
1.050
1.045
1.040
1.035
1.030
1.025

5 Supporting data

The Supplementary Materials provide consolidated run-level outcomes and machine-readable summaries for the rankings, independent-sample tests, effect sizes, runtime estimates, granularity profiles, and controller trajectories. RQ3 is descriptive and does not estimate a factorial interaction.